

**WEB BASED TEA GROWER
SUPPORT SYSTEM
FOR
TEA RESEARCH INSTITUTE OF
SRI LANKA**

U D ALAGIYAWADU

February 2021



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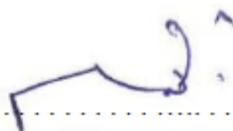
**This dissertation is submitted in partial fulfilment of the requirement of the
Degree of Bachelor of Information Technology (external) of the
University of Colombo School of Computing**

Declaration

Declaration

I certify that this dissertation does not incorporate, without acknowledgement, any material previously submitted for a degree or diploma in any university and to the best of my knowledge and belief, it does not contain any material previously published or written by another person or myself except where due reference is made in the text. I also hereby give consent for my dissertation, if accepted, to be made available for photocopying and for interlibrary loans, and for the title and abstract to be made available to outside organizations.

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Abstract

Tea Research Institute of Sri Lanka is the main national organization that generates and disseminates scientific knowledge and technologies necessary for tea industry in Sri Lanka. Since 1925, TRISL has contributed to the industry through its continued commitment. Here, several information channels such as printed media, training programmes, seminars, audio visual materials and mass media techniques are used in order to disseminate information as recommendations to tea growers. Further, the TRISL website has provided primary information and recommendations to the public. In addition, field visits are conducted by the researchers and extensionists to solve the field related problems of tea growers.

At present, client maintaining records manually in the record books or spreadsheet and word processing packages i.e. MS Excel, MS Word and also arrangements were done manually. However, there are lot of redundant records and difficult to find data as and when necessary. Therefore, the primary objective of this project is to develop a system to offer services online facilitating tea growers. Further, it will provide avenues for client to update novel scientific knowledge and technologies as and when available and applicable. This is a versatile system which handle online member registration, manage tea land data, facilitate land selection for tea cultivation, booking for soil sample analysis, diagnosing pest and disease damages and providing necessary control measures.

The system based on client-server architecture as well as non-object-oriented iterative software development methodology, which known as Rational Unified Process is used for modeling. The Unified Modeling Language is used to design the system. A server-side scripting language of PHP with HTML, JavaScript and CSS are used for front-end development. Apache NetBeans IDE is used as integrated development environment. The Apache is used as the web server and the MySQL is used to handle the database. Since this is a web-based system, it facilitates to use any GUI based OS platform with a web browser. This system has proposed to fulfill the client's requirement. It will really help TRISL to ease their technology dissemination activities and deliver e-services more productively and well-organized manner to the public.

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List of Acronyms

AER	- Agriculture Ecological Region
BIT	- Bachelor in Information Technology
CEO	- Chief Executive Officer
CRM	- Customer Relationship Management
CSS	- Cascading Style Sheets
DB	- Database
DOA	- Department of Agriculture
DS	- District Secretariat
ERD	- Entity Relationship Diagram
FOSS	- Free and Open Source
GN	- Grama Niladhari
GPL	- General Public License
GUI	- Graphical User Interface
HTML	- Hypertext Markup Language
IDE	- Integrated Development Environment
JS	- JavaScript
NRMC	- Natural Resources Management Center
PHP	- Hypertext Pre-processor
RDBMS	- Relational Database Management System
RUP	- Rational Unified Process
SQL	- Structured Query Language
MS	- Microsoft
OS	- Operating System
SDLC	- System Development Life Cycle
SLTB	- Sri Lanka Tea Board
SPN	- Soils and Plant Nutrition
TGSS	- Tea Grower Support System
TRISL	- Tea Research Institute of Sri Lanka
UI	- User Interface
UML	- Unified Modelling Language

Chapter 1 – Introduction

1.1 About the Client

Tea Research Institute of Sri Lanka (hereafter referred to as TRISL, TRI or client) is the main national organization that generates and disseminates scientific knowledge and technologies necessary for improving and sustaining tea cultivation and processing in Sri Lanka. Since its establishment in 1925, TRISL has contributed to the tea plantation sector through its continued commitment as is highlighted in their mission statement, as follows;

“To generate and transfer scientific knowledge and technologies appropriate for the stakeholders to improve productivity and quality of Sri Lankan tea in a profitable manner” [1]

1.2 Motivation for the Project

The client uses several information channels such as printed media (i.e. text books, advisory circulars, leaflets, guidelines etc.), conducting trainings, seminars, audio visual materials, telephone calls and mass media techniques (i.e. radio, television) for disseminating information. However, these channels are ineffective and inadequate to meet the urgent, specific and enormous information needs of tea growers as well as other stakeholders in the tea sector. Moreover, such communication channels proved that time consuming and poor in reaching large number of audience within a short time effectively. In addition, growers cannot reach easily to the information when necessary through above channels.

Tea growers would certainly require accessing of latest technologies developed by TRISL and updated information may be within a shorter time, to facilitate their decision-making process.

Therefore, it is a responsibility of TRISL to develop such a system to fulfill this requirement incorporating TRISL recommendations. Although TRISL has its own website [2] which contains primary information and recommendations generated by the researchers and extensionists does not facilitates above said kind of services. Hence, TRISL needs some sort of efficient and effective way to fulfill this. When facing kind

of a pandemic situation i.e. recent COVID-19 outbreak, it would be more useful solution to reach stakeholders of TRISL.

In this digital age in order to overcome the inefficiency of information channels, can be introduce excellent solution using online information service portal which link the TRISL and stakeholders. According to my belief, this kind of a system would help the client to carry out effective technology dissemination.

Moreover, this attempt would greatly help to improve my skills on project management, system analysis, designing and development while completing my studies leading to the Degree of Bachelor in Information Technology.

1.3 Objectives the Project

The objectives of the project are listed below;

- Develop an online system to manage tea information facilitating tea growers
- Provide avenues for client to update the novel scientific knowledge and technologies when available and applicable
- Reach more tea growers and their management agents to disseminate scientific knowledge and new technologies within a short time efficiently and effectively to facilitate their decision-making process
- Facilitate generating service requests easily
- Speeding up the operation
- Reduce data redundancy
- Secure data storage for authority end
- Practically apply knowledge and skills gained throughout the degree programme

1.4 Scope of the Project

The scope of the project listed as follows;

- Keep grower and tea land records in digitally accessible format
- Manage member and tea land registrations
- Receive requests from tea growers such as (a) Selecting suitable lands for tea growing, (b) Booking soil sample analysis, (c) Diagnosing of disease, pest damages etc of tea fields and manage them

- Recommend latest tea cultivar/s for new cultivations based on Agriculture Ecological Regions
- Keep records in nursery plants and tea shoots availability of mother bush sites in respective regional areas of Sri Lanka
- Manage samples, analysts, laboratories and also ensure payments that receiving prior to start analysis
- Direct diagnosis request efficiently to Subject Matter Specialists to prepare reports with recommendations and deliver them to members
- Facilitate to send email alerts to the manager/proprietor for enabling them to apply correct management practices to their tea fields
- Provide authenticated user login to access the system securely for the users
- Module wise user and access privilege management
- Validate user inputs
- Provide management reports (i.e. details of registered members, member activity reports for given period, no of analyzed samples in given period, details of pest and disease occurrences, etc) to the management to facilitate decision making

1.5 Outline of the Dissertation

This dissertation contains six main chapters, as Introduction chapter describes flow of development including overall view on this online system. Given below is the outline of succeeding chapters.

1.5.1 Chapter 1 – Introduction

This chapter briefly describes about the client, current manual system and its drawbacks. Also includes motivation for the project, objectives and its scope. Further, it introduces structure of the dissertation.

1.5.2 Chapter 2 – Analysis

This chapter explains about the details of existing similar systems, requirement gathering techniques, collected functional and non-functional requirements to design and development of the system.

1.5.2 Chapter 3 – Design

This chapter provides the design of the proposed system together with necessary design documentation. It also shows in detail how the user interfaces, modules and database of the system.

1.5.3 Chapter 4 – Implementation

The implementation chapter explains the development process of the system. Further, the implementation environment, software tools, technologies used at the time of development and implementation in the client domain are explained. This chapter also incorporates module structure, major code and reused modules of the system.

1.5.5 Chapter 5 – Evaluation

The test plan and test cases of the system describing testing process that given in this chapter. It identifies whether all aspects of the system are working correctly.

1.5.6 Chapter 6 – Conclusion

Here, a critical discussion and assessment of results of the project is discussed and also lists problems encountered, lessons learnt and possible further developments.

After these chapters, in a Reference section all the materials referred to develop this system and write the dissertation included. Moreover, in the Appendices, System Documentation, Design Documentation, User Documentation, Management Reports, Test Results, etc are provided.

Finally, a Glossary of Terms and General Index are included.

Chapter 2 – Analysis

The requirement analysis is one of the most important phases of a system development project. Therefore, significant amount of time and effort was spent in this segment. The main processes of this phase were involved domain understanding, requirement gathering, classification, structuring and prioritization. The software development industry has learned that in order to successfully complete an information system, the system analyst first must understand the need of the stakeholders and the reason to develop the system – a concept called “user-centric development”.

2.1 Fact Gathering Techniques

Out of many fact gathering techniques available for the system analysis, followings were used to capture requirements to obtain thorough knowledge of domain understanding.

1. Interviews
2. Observation of work environment
3. Examining existing documentation

2.1.1 Interviews

However, this system is quite a novel concept for the client and the tea growers, the main requirements were gathered by interviewing the Director of TRISL. Furthermore, Deputy Director (Extension), Deputy Director Research (Production) and following sectional heads were also interviewed to gather additional knowledge needed to complete the analysis.

1. Head, Advisory and Extension Services Division
2. Head, Plant Breeding Division
3. Head, Entomology and Nematology Division
4. Head, Plant Pathology Division
5. Head, Soils and Plant Nutrition Division
6. Superintendent, St. Coombs Estate

When growers requesting to assess suability of tea lands, recommend best suited tea cultivars, analyze soil samples, identify pest damages and diagnose diseases etc of their tea lands, the client is available to perform their requirements. Therefore, relevant subject specialist researchers were identified to provide relevant requirements. Above

interviews were conducted as open-ended and at the first instance 20-45 minutes duration was spent per person.

Since proposed system that requires data related to tea fields (i.e. planted date, type of plants, cultivar name(s), pruning date, extent, etc) and other necessary data of tea lands, most appropriate information source would be administration-end of tea estates. Therefore, St. Coombs estate, which is under the purview of TRISL was selected to discover related requirements mentioned above.

2.1.2 Observation of work environment

Observation allows the system analyst to study what information useful for users to carry out their job. It becomes beneficial, when we intend to acquire more knowledge and understanding about the domain.

2.1.3 Examining documentation

Relevant documents and paperwork of the existing manual systems were analysed through a sampling process. Solid information could capture mostly by performing this technique and facilitate to validate gathered requirements by other gathering techniques.

2.2 Existing Manual System

Physical structure of the TRISL consisting with the Head Office, six Regional and Extension Centers and four Regional Soils and Plant Nutrition Laboratories.

Directly associated staff of the system is Head, Advisory and Extension Services, Regional Heads, Advisory/Extension Officers (Field Officers), Subject Matter Specialists, Head, Soil & Plant Nutrition (Laboratory in Charge), Laboratory Analysts and Cashiers.

2.2.1 Functionalities of existing system

TRISL has been following a manual approach based on papers and requests were received by snail mail, fax or email communications from tea growers. Therefore, during this phase it was identified that less functionality was performed over the years by the current system such as:

- Maintain records of tea growers as MS Word and MS Excel documents.
- Maintain grower requests (paper based printed format) in a manual filing system.
- Sample analysis bookings were made over the phone occasionally and recorded in a register manually.

- Payment records for sample analysis were maintained in a book manually.
- List of tea nurseries and plant availability are recorded as MS Excel documents.
- List of regional sites of tea mother bush and shoot availability with available time periods were recorded as MS Word and MS Excel documents.
- Land suitability parameters, List of recommended tea cultivars for different regions etc need to be obtained by referring paper based TRISL Advisory Circulars.
- When a land suitability request for new tea cultivation forwarded by a tea grower,
 - (a) If the request received related to Corporate tea land (estate), respective Regional Head will be contacted by the Estate Manager (Superintendent) over the phone and visit the location
 - (b) If the request received related to private tea land, respective Regional Head will be communicated over the phone or by physically meeting Tea Inspector of the region attached to Tea Small Holding Authority (TSHDA), contacted the land owner and visit the location

During the above process, requests were received by snail mail, fax or email to the client. Those requests were entered and maintained in book manually and/or maintained in an office file.
- Relevant reports were also forwarded to tea growers by snail mail. Sometimes scanned reports were converted and emailed growers.

The basic workflow of the existing system is shown below in Figure 2.1

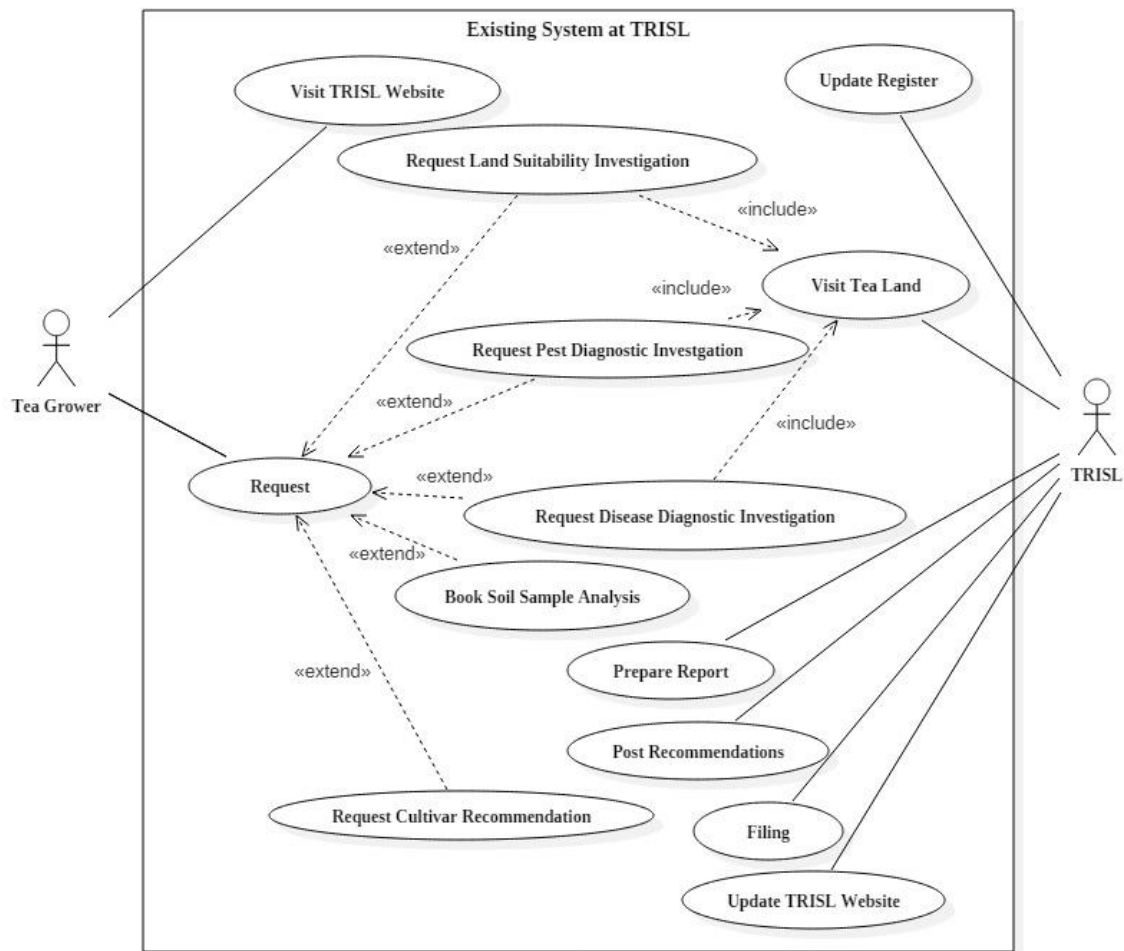


Figure 2. 1 Top level use case diagram of the existing system

2.2.2 Drawbacks of existing system

The existing manual system has following major drawbacks listed below.

- There is no complete and organized list of tea growers.
- Handling requests of land suitability assessments for new tea cultivation and field diagnosis for pest and disease damages were done manually. Therefore, assigning officers to land visits, prepare recommendation reports and finding previous reports were tedious.
- Performing manual booking procedure and allocating laboratory analysts for soil sample analysis are very difficult specially in peak periods.
- Maintaining up to date plant availability of nurseries and shoot availability of tea mother bush sites were also tedious tasks.
- Possibility of data redundancy is high.
- Data retrieval is time consuming and taking lot of effort to complete.

- Experiencing data losses due to misplacing recorded materials.
- Lack of data security.

2.3 Functional Requirements

In Software Engineering, a functional requirement defines a function of a system or its component. A function is described as a set of inputs, behavior and outputs. Functional requirements may be calculations, technical details, data manipulation, processing and other specific functionality that define what a system is supposed to accomplish [3]. An in-depth study was carried out in order to understand the full functionalities of each process of manual system. The system has been divided into 8 modules in order to get a clear picture of the main processes. Identified functional requirements detailed as follows.

2.3.1 Manage Tea Land Information

- Allow to register tea lands by obtaining the Register of Tea Estates/Lands of the Tea Commissioner of Sri Lanka, since the Tea Commissioner maintains prescribed particulars of tea lands, their management agents and proprietors [4].
- When visitor (tea grower) trying to register as a member, details of respective tea land should be selected from the system. Therefore, the database need to be updated with name of the tea land, Sri Lanka Tea Board registration number, management agent, land ownership category, administrative district, DS division, GN division, village name etc.
- The member registered for the respective tea land as the delegate, need to update following details at first instant.
 - ✓ Details of divisions
 - ✓ Details of tea fields of respective divisions

2.3.2 Manage Tea Grower Information

- All the facilities will be provided only to registered members. Therefore, a visitor to the system should be facilitated to register.
- Once the visitor fills and submits the registration form there should be a method to inform the Administrator.
- The client should have an authorization procedure to permit respective visitor to selected tea land when requesting member registration.

- A registered member required to logon to the system using his or her account, insert and modify details as and when necessary.
- A password recovery procedure will also be provided in case member forgets the password.

2.3.3 Marge Land Suitability Assessment Requests

There are many factors that determine land suitability for tea cultivation. Therefore, it is necessary to propose a method to obtain simplified land suitability classification criteria based on limitation factors included in following tables (Table 2.1, 2.2 and 2.3) [5].

Climate (agro-ecological region)	Limiting factor				Degree of limitation
	Slope (%)	Soil depth (cm)	Surface rockiness (%)	Gravel (Volume % in top 50cm)	
					Regions above 900m elevation
WU2, WU3, IU2 & IU3	0-25	>90	0-10	0-10	None
IU1	25-70	70-90	10-20	10-50	Moderate
All other up country regions	>70	<70	>20	>50	Severe
					Regions between 300 - 900m elevation
WM3	0-25	>90	0-10	0-10	None
WM2 & IM2	25-55	70-90	10-20	10-50	Moderate
All other mid country regions	>55	<70	>20	>50	Severe
					Regions between 300m elevation
WL1	0-25	>90	0-10	0-10	None
WL2	25-70	70-90	10-20	10-50	Moderate
All other low country regions	>70	<70	>20	>50	Severe

Table 2. 1 Criteria for classifying lands according to their suitability for tea – single limitations

Moderate limitations	Climate	Slope	Soil depth	Surface rockiness	Gravel
Climate	Class 2	Class 3	Class 4	Class 3	Class 3
Slop	Class 3	Class 2	Class 4	Class 3	Class 3
Soil depth	Class 4	Class 4	Class 2	Class 3	Class 4
Surface rockiness	Class 3	Class 3	Class 3	Class 3	Class 3
Gravel	Class 3	Class 3	Class 4	Class 3	Class 2

Table 2. 2 Criteria for classifying lands according to their suitability for tea – dual moderate limitations

Suitability Class	Degree of Limitation	Suitability
Class 1	No significant limitation	Highly suitable
Class 2	Moderate single limitation	Suitable
Class 3	Moderate dual limitation	Moderately suitable
Class 4	One or more severe limitations, strongly interacting dual limitations or multiple moderate limitations	Unsuitable

Table 2. 3 Land suitability classification based on limitations

After classifying using above physical features of the land, it is necessary to consider crop suitability recommendations according to the Grama Niladhari (GN) Division (the smallest administrative unit in Sri Lanka) level. Suitability information relevant to different agro-ecological regions for the tea cultivation were extracted from Hand Book on Tea [7] and online tool which was developed by the Natural Resources Management Center, Department of Agriculture, Sri Lanka [6]. This system needs to facilitates to select the land considering agro-ecological regional variation at the GN Division where land is located.

2.3.4 Mange Cultivar Recommendation Requests

- Support managing suitability of tea cultivars for the different climatic regions.
- Support inserting new cultivars, view and modify existing cultivar details including according to different characteristics.
- Facilitate to select at least 5 - 6 cultivars depending on their suitability for the particular location and extent.
- The recommended lists of cultivars need to obtain from TRISL Advisory Circular No. PN01 [8]. The “Recommended cultivars for planting in different climatic regions” and “Characteristics of recommended cultivars” were given therein.
- Facilitate updating of new cultivar releases, remove unsuitable cultivars and modify characteristics of existing cultivars.

2.3.5 Mange Sample Analysis Requests

- The system should allow managing sample analysis booking requests prior to handover soil samples to TRISL Regional Laboratories by tea growers.
- The system should allow maintaining payment details up to date.
- It should facilitates viewing and downloading facility of analysis reports.
- The system should allow facilitating of managing laboratories.

- Assigning an analysis to a laboratory and an analyst to a laboratory also need to be done.

2.3.6 Manage Disease and Pest Diagnostic Requests

- This facility supports to manage diagnostic service requests to identify plant diseases and pest damages to the tea fields.
- To facilitate this photograph submission facility for the members will be provided.
- Once submission receives to the system there should be a method to inform identified subject matter specialist.
- All the details and photographs should be examined by the most appropriate subject specialist and he/she will need to reply by preparing diagnostic report.
- The system should facilitates viewing and downloading facility of diagnostic reports.

2.3.7 User and Module Management

- This module needs to provide system administrator to create and manage user accounts for employees of the client.
- The details of members need to be managed.
- Should facilitate managing access permission to system modules accordingly.
- This should also include the facility to activate and deactivate a member account of tea grower.

2.3.8 Report Generation

- The system must provide facilities to generate day to day transaction reports (i.e. Invoice for Sample Analysis, Sample Analysis, Land suitability assessment, Number of members registered in a day, Availability of tea shoots, Availability of tea plants, etc.) as well as summarized reports for the management (i.e. Total number of Members registered over the time, Total number of diagnosis requests over the time etc.)
- Support to generate reports in pdf etc formats
- Allowing user to print selected reports

2.4 Non-functional Requirements

According to requirements followings were identified as non-functional requirements of the proposed system. They specify criteria that judge the operation of the system rather than specific behaviors. They are generally invisible to the users.

2.4.1 Availability

Since part of the proposed system is a website this will be a key requirement and tea growers are willing to access it as their convenient time and location. It is important to have 99.9% up and running time of the hosted server to keep the system available and operational on 24x7.

2.4.2 Portability

If a server migration or a release changes, software platform need to be done to the system, should be able to handle without much difficulty. Furthermore, release dependent coding which are not backward compatible to a release and inbuilt code functions expected to be deprecated have to be avoided.

2.4.3 Response and Processing times

Nielsen [9] outlined basic response time recommendations based on the way human attention work. After 10 seconds user attention is completely get lost according to him. Usually, 40% of users will abandon a website after 3 seconds. Therefore, the system should respond to user action as soon the request generated. Therefore, all possible efforts need to be implemented to reduce the wait time of the user for a specific operation to happen and response times should be as fast as possible. Targeted users, specially tea growers will certainly appreciate about said performance of the system.

2.4.4 Reliability

Based on research inputs of land suitability assessment parameters and constituent levels of samples to be analysed are determining. Therefore, 100% accuracy is needed when making recommendations. Accurate information which will be generated while functioning the system for long period could be used to generate new research findings.

2.4.5 Security

The system is handling important and sensitive data related to business processes of tea growers. Therefore, no unauthorized access to the system should be guaranteed. Timely

backups should be taken to keep the reliability and appropriate security measures should be adhered to limit accessing unwanted persons to protect data.

2.4.6 User friendliness

The system should provide user-friendly atmosphere to scroll up and down the interfaces, where appropriate menus and links were positioned in suitable locations. Majority of the UI design rules were also needed to consider when designing as well as providing clear and consistent design, helps in minimizing the number of clicks. A user would perform in navigating from one place to another, use images and icons to make the user feel aware what he/she willing to do, users prefer interfaces are simpler and using without learning.

2.4.7 Usability

One of the major requirement of a software system or website is usability. Ease of use of system/website earns more users. Easy to adapt with simple navigation supported user interface design together with consistent look and feel should be maintained in this regard. Another concern for usability is ensuring that the website is appropriate for all ages, genders and levels of tea growers. Easy to understand error messaging should be provided as and when applicable. The targeted users of the system are familiar with paper based system and some basic computer applications (i.e. word-processing and spreadsheet). Therefore, backend of the system should be easy to use even for an officer who does not have much technical knowledge. Scientific terms related to tea will be using in the system, so it will become more familiar to the users.

2.5 Existing Similar Systems

As described in the “Introduction” Chapter this is a less explored field of information usage for tea growers in Sri Lanka. Hence, finding a similar Sri Lankan system or website was a tedious task.

There are few numbers of online systems developed by other countries that support some selected functionalities to some extent but serving for their local requirements and cannot be compared to the objective of this system at all. The system feature comparison is given in Table 2.4.

2.5.1 Farmers' Portal

The Farmers' Portal (one stop shop for farmers) which was developed by Department of Agriculture and Cooperation (DAC) attached to Government of India is an online web portal envisage to making available relevant information and services to the farming community. A screenshot of the portal is shown in Figure 2.2. A farmer will be able to get all relevant information on specific subject around his village, district or state. Also farmer will be facilitated to ask specific queries as well as give valuable feedback through the “Feedback module”. It includes Identification of agriculture lands using country map, Soil Health Cards of different areas, Disease and symptom details etc. In addition, there are many number of information relevant to Agriculture and Animal Husbandry are available in the system [10]. It is available to access on the web in <https://farmer.gov.in/FarmerHome.aspx>. This system supports following major crops such as Wheat, Maize, Rice, Pulses, Jute, and Sugarcane etc and not intend to use of Tea.



Figure 2. 2 Screenshot of Farmers' Portal

2.5.2 Soil Test Pro

The Soil Test Pro is complete online soil nutrient management system which is available at <https://www.soiltestpro.com> [11]. This system has two primary components as Web based portal (named “Web Headquarters”) and Mobile App, both are free to access. A screenshot of the portal is shown in Figure 2.3. It facilitates to register as a farmer in their server and then the user needs to setting up farm and its field by marking boundaries using online map feature. Then the user will be directed to create soil sampling job and the system facilitates to select recommended lab from the list to ship the samples. Their specialists will notify the farmer when lab results have been updated to the system within shorter period. The Soil Test Pro is product of TapLogic LLC based in Kentucky, United States. System is not specific to any crop but their labs are located in United States and analysis are based on their recommendations.

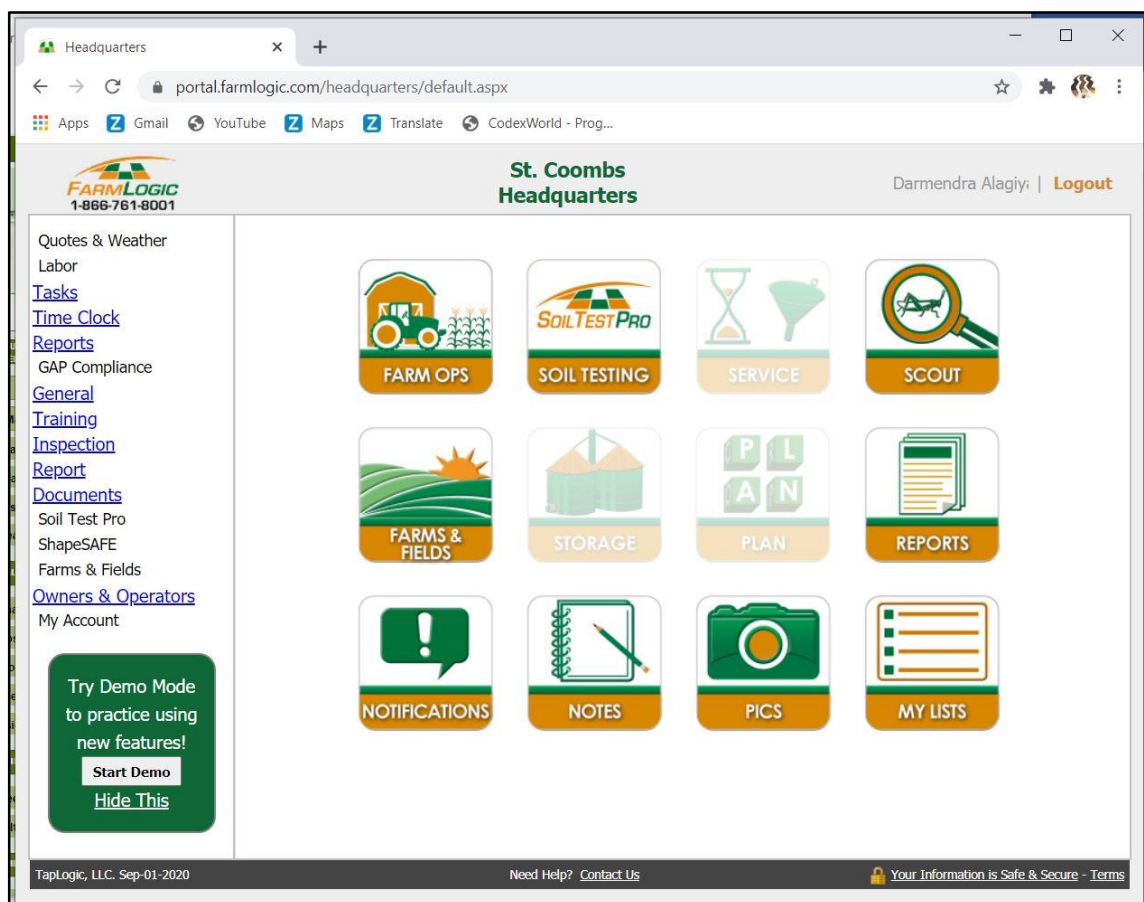


Figure 2. 3 Screenshots of Soil Test Pro home page

2.5.3 Plantix Version 3.3.0

Plantix 3.3.0 is an Android App and can be used to detect pests and diseases on crops within seconds. Plantix serves as a complete solution for crop production and management. When farmer take the picture of disordered plant and uploaded through the Plantix, it detects diseases and deficiencies followed by recommendations on most effective treatments [12]. Moreover, it offers Fertilizer Calculator, Weather Forecast and Cultivation Tips. It covers 30 major crops and detect more than 400 plant damages supports to 18 languages. However, no support available for Tea.

2.5.4 CropDiagnosis Version 1.1.0

CropDiagnosis 1.1.0 is also an Android App aiming to improve pest management decisions by making crop diagnosis more accurate, selection of chemicals error-free and application assisted by personalized instructions. By evaluating the crop's details (type, location, soil, history, etc) and threat's characteristics (type, appearance, progress, etc), as entered by the user through a smart questionnaire, the app suggests the most likely diagnosis [13]. This App supports crops such as Cotton, Cereals, Pepper, Lemon, Olive and Tomato but not includes Tea like perennial crops.

There are some websites and blogs which would occasionally publish chosen contents to some extent, but serve for different purpose and cannot be compared to the objective of this system. Hence, no exactly similar commercial or open source project was discovered during the review done. The comparison between proposed system and above similar systems are shown below in Table 2.4.

Feature	Proposed TGSS System	Farmers' Portal (India)	Soil Test Pro (United States)
Supported Crop	Tea	Wheat, Maize, Rice, Pulses, Jute & Sugarcane	Any Crop
Web Presence	Yes (web based system)	Yes (online tool)	Yes (online tool)
Android App	X	X	√
Online	√	√	√
Grower Registration	√ (multiple)	X	√ (multiple from server)
Land/Farm Registration	√ (multiple)	X	√ (multiple from server)
Field Data Management	√	X	√
Crop-Land Suitability Check	√	√	X
Manage Sample Analysis Requests	√	X	√
Disease and Pest Diagnosis	√	X	X
User Management	√	X	√ (on the server)

Table 2. 4 Similar System Comparison

2.6 Process Model for Proposed System

Researchers and practitioners have identified a number of models in software development process [14] such as Waterfall, Agile, Scrum, Rational Unified Process (RUP) and Rapid Application Development (RAD).

Here, Waterfall method was used for modeling and is the earliest SDLC approach and referred as a linear-sequential model. Hence, requirements of this project were well defined with limited scope it was selected. The selected model allows to understand that further levels are made progressive upon the completion of a previous one. Each phase consists on a definite set of activities and deliverables that must be accomplished before the following phase can begin.

The phases are always named differently but the basic idea is that the first phase tries to capture *What* the system will do, its system and software requirements, the second phase determines *How* it will be designed. The third stage is *where* the developers start writing the code, the fourth phase is the *Testing* of the system and the final phase is focused on Implementation tasks such as training and heavy documentation [15]. Figure 2.4 shows a traditional waterfall lifecycle.

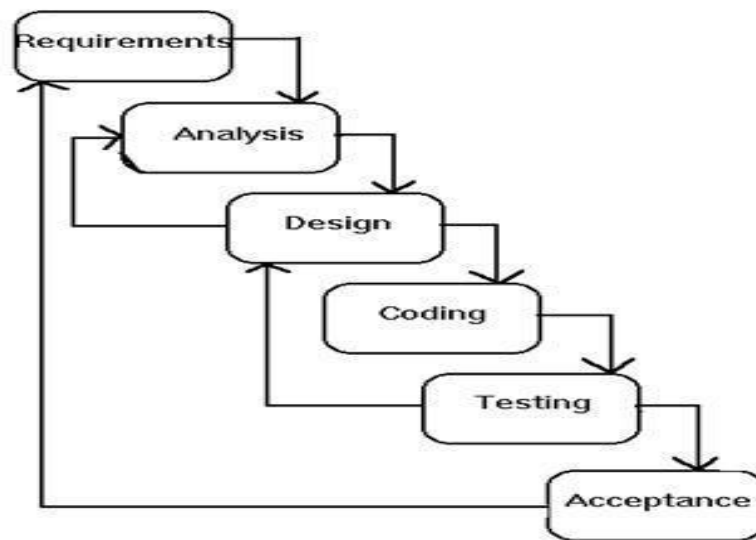


Figure 2. 4 Traditional waterfall lifecycle [15]

The RUP is an iterative model which enhances the evolving versions until the full system is implemented [14]. Because each RUP iteration comes out with an executable release based on client feedback, the client gets realized benefits much earlier than Waterfall model. Aim is to develop the project with limited time scope and to meet the PEB requirements in mind RUP was not used. Since Agile method is not selected as it is well suited for a project that is having vague requirements whilst the requirements of this project were well defined. The Agile method is more suitable for software projects that have rapidly changing requirements [16].

Chapter 3 – Design

The overall structure of the proposed system was based on the design chapter. Design methodologies, tools and techniques which were used in the design phase and the database of the system described in this chapter along with suitable UML diagrams.

3.1 Way of Development

The system was planned to develop using open-source components. Since open-source technologies are less-costlier and client is a middle-size enterprise, it would be a better selection. A system development from scratch is more time consuming though custom built solutions are very accurate.

3.2 Development Platform

It was decided to develop web based system and reasons for the selection is described in section 3.2.4 in detail.

3.2.1 Alternate Solutions

Alternate solutions for a web-based system are Standalone systems, Mobile Applications, collection of software (i.e. Customer Relationship Management (CRM), Lab Report Management, Invoice Management software etc) or practicing current manual system.

3.2.2 Standalone Software

Standalone software system does not bundle with set of separate applications. Such software does not require an internet connection to perform its functionalities. This kind of support system cannot implement as a standalone system. Because it is not match with the primary objective of this project is to address problem of inefficiency of information channels that the client currently uses.

3.2.2 Collection Software Applications

Following set of software applications are used to manage some facilities of problem domain.

- CRM software for grower and tea land profile management
- Lab report management software to handle sample analysis
- Invoice management software to manage charges of sample analysis

- Report generation software to generate sample analysis reports, invoices, payment receipts and other management reports

There are some disadvantages of using separate software for different tasks such as more expensive, have to pay for unnecessary features, heavy workload for the staff, involvement of data and work repetition.

3.2.3 Reasons for Choosing Web based System

Most suitable method of attracting the user base is web and it allow accessing the system for more number of users at any given time. Following are the reasons;

- Platform independent - system would be platform independent and hence tea grower can easily access the system from any computer connected to the internet
- Cost effective development - a single, responsive web application need to test thoroughly on different web browsers, no need to test the application on different browsers. This makes development and testing much easier. It cuts development costs and reduces development time.
- Easy customizable - client interface will run on the web browser. Therefore, dependencies relevance to client machine is minimum.
- Accessible anytime and anywhere - unlike standalone applications, web based systems are accessible anytime, anywhere and leading to the problem of inefficiency of traditional information channels which are unable to reach the grower within the short time.
- Increased security features - since application is hosted on a dedicated server it is more effective than implementing and monitoring security features [18].
- Improved processing time - loading time on the client machine is lesser than standalone, because the user is simply redirected to the appropriate page.

3.3 Architectural Design of the System

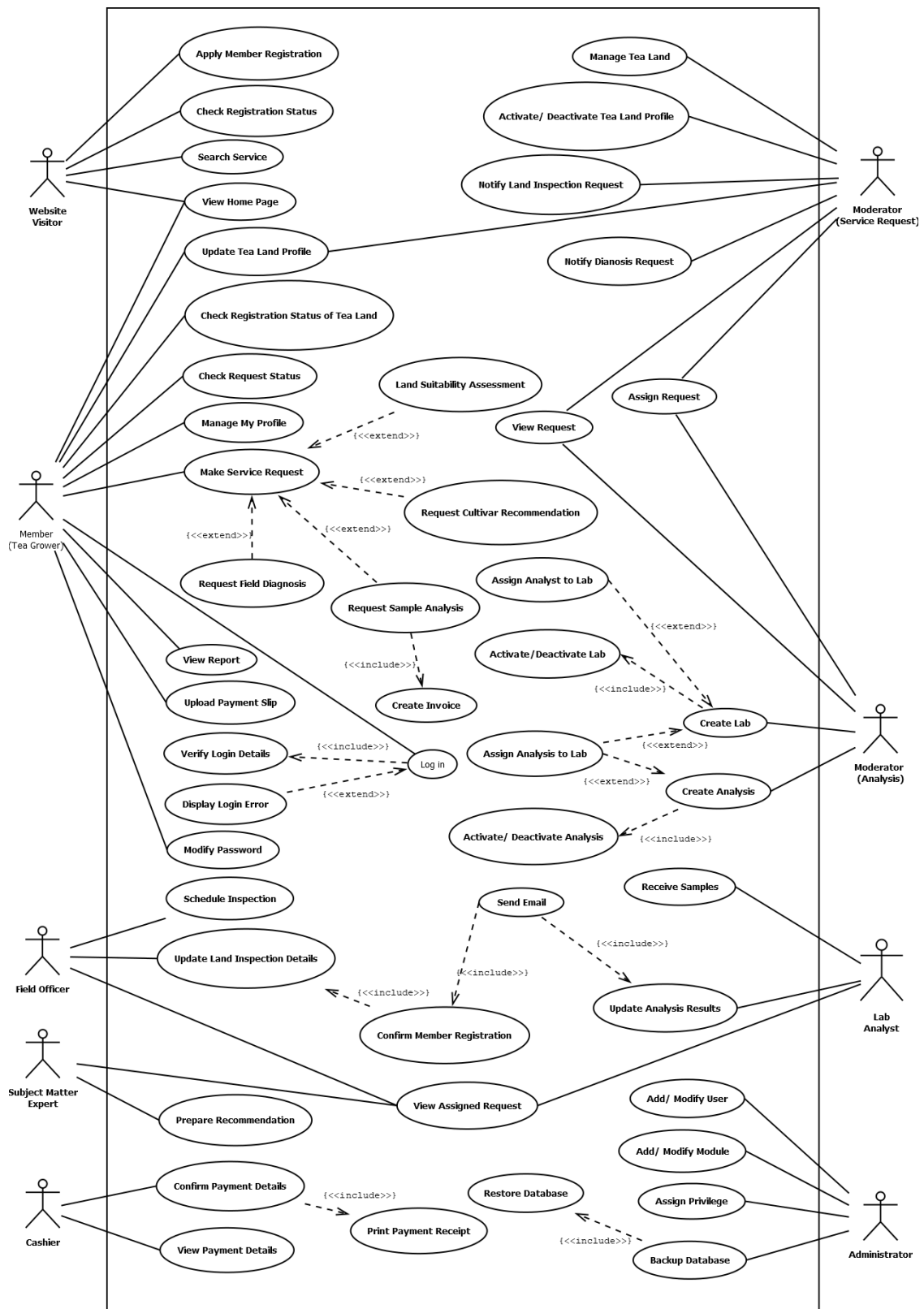
The system is planned to develop using Non-Object Oriented as a Procedural Programming concept. The Unified Modeling Language (UML) is used for the graphical representation and documentation of the design. The UML is a standard language for specifying, visualizing, constructing, and documenting the artifacts of software systems [17]. UML diagrams are the ultimate output of the design phase, which consist of use case diagrams, use case narratives, activity diagrams and entity relationship diagrams.

3.4 Proposed Design

3.4.1 Use Case Diagrams

A use case diagram graphically describes who use the system and how a user expects to interact with the system. Figure 3.1 shows the high-level use case diagram which represents the proposed system. It includes actors, use cases and relationships among use cases and actors. Identified use cases represent refined functionalities of the system. The followings were identified as actors of the proposed system.

- Website visitor (Applicant): A tea grower who is intend to obtain a service from TRI
- Member: A tea grower who joined the system
- Moderator: An employee who is assigned to manage member and tea land registrations, assigning diagnosing requests to subject matter experts etc
- Filed Officer: An employee who is evaluating land suitability requests and tea cultivar recommendation
- Subject Matter Expert: An employee who involved in diagnosing pest or disease by studying member request
- Lab in Charge: An employee who is managing regional sample analysis laboratories, analyses, analysts and analysis reports
- Laboratory Analyst: An employee assigned to perform sample analysis and create analysis reports
- Cashier: An employee assigned to receive payments and update the status
- Administrator (Amin): An employee assigned to manage system users, modules, making system backups and restoring them



The suggested solution has been divided in to 8 modules in order to make development systematic and maintain easier. Individual use case diagrams, which were prepared to

represent the conceptual model, were included to illustrates them simply and more meaningfully with the actor(s).

1. User and Module Management

Handling users is very much important part of a software system. This module allows administrator to add new user, modify relevant user details, change user status and related activities. Also this module facilitates assigning roles and make changes to them will be handled. Figure 3.2 is depicted the use case diagram for this module.

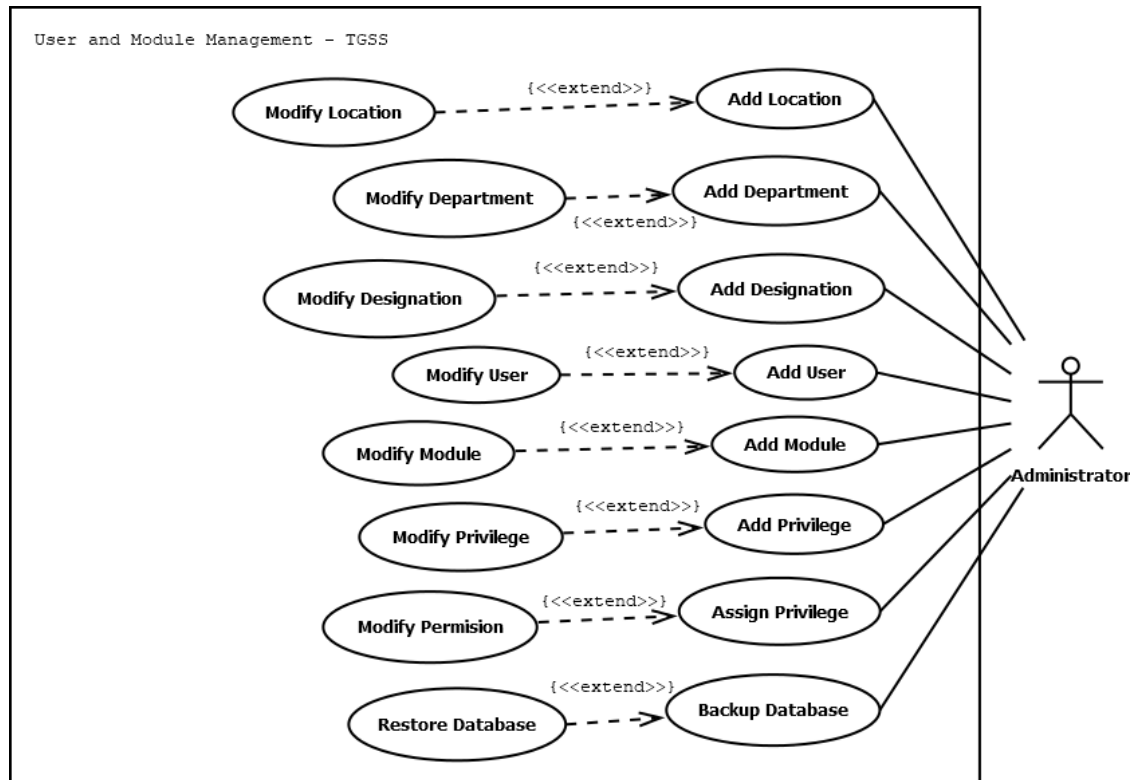


Figure 3. 2 Use case diagram - User and Module management

2. Member Registration Module

The use case diagram for Member Registration module is given in Figure 3.3 and illustrates how the actors are involving during member registration.

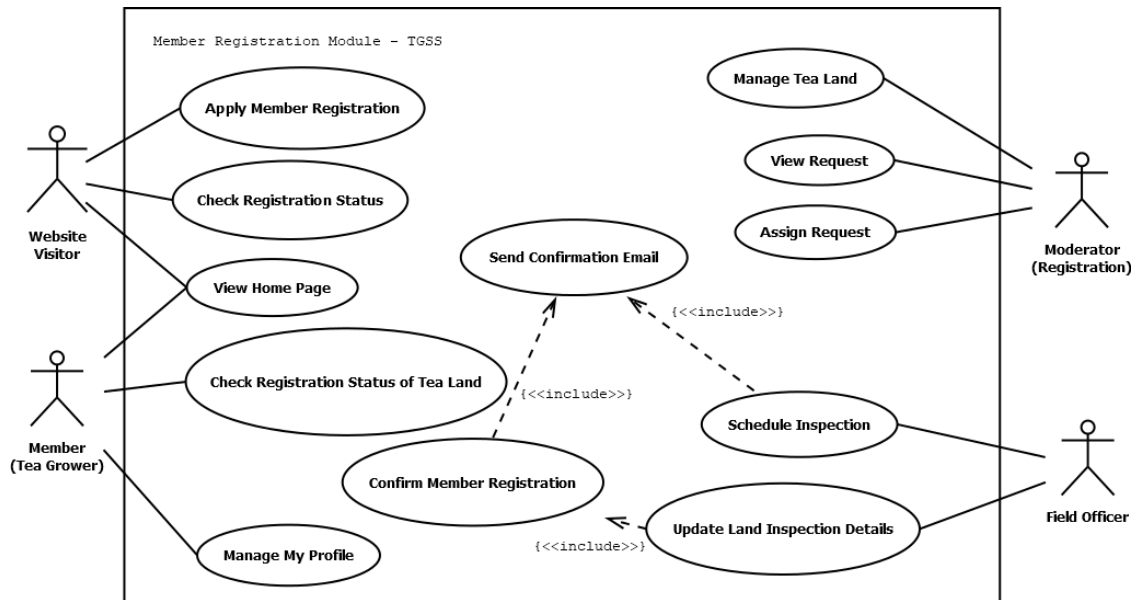


Figure 3. 3 Use case diagram - Member Registration module

3. Land Suitability Assessment Module

Figure 3.4 depicts the use case diagram of Land Suitability Assessment module.

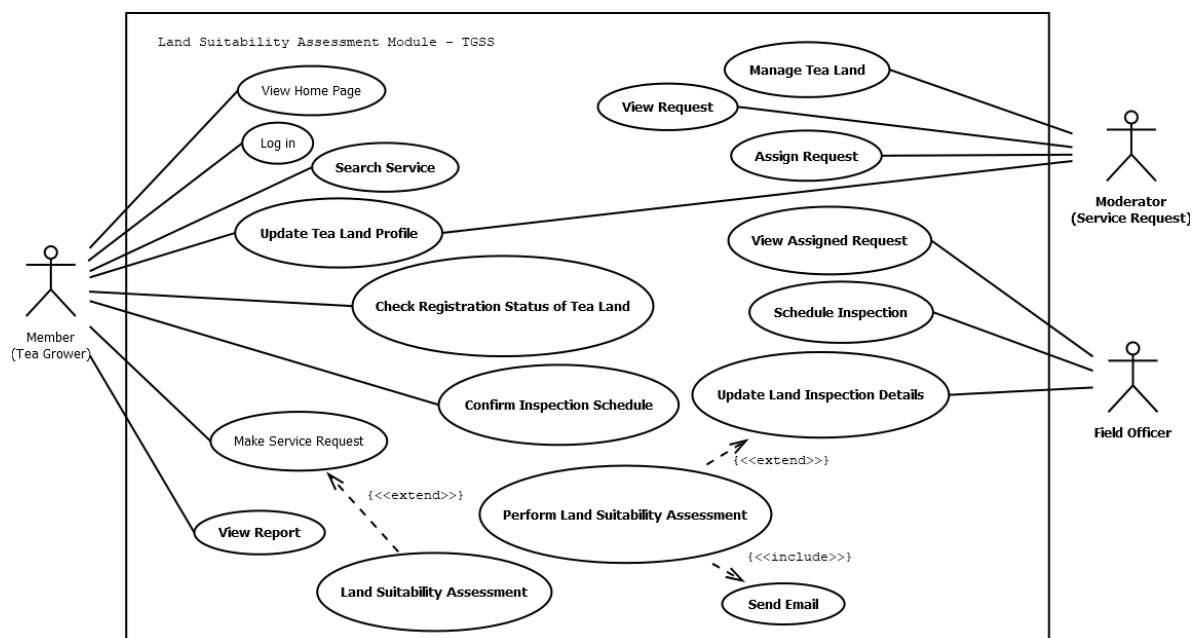


Figure 3. 4 Use case diagram - Land Suitability Assessment module

Rest of the use case diagrams which were prepared to represent the conceptual model are included in the Appendix B.

3.4.2 Use Case Narratives

Some of the main use case narratives are depicted below in Table 3.1 and 3.2 according to the use case diagrams. The rest of the narratives are included in Appendix B.

Use-case Number	UC-01	
Use-case Name	Login	
Priority	High	
Actor/s	User	
Description	This use-case describes how actors log into the Tea Grower Support System	
Precondition	None	
Post condition	If the use-case was successful, the actor logged into the system. If not, the system state is unchanged.	
Basic course of action	User Action	System Response
	1. The actor visits home page to login to the system. 3. The actor enters username and password then click on Login button.	2. The system prompts the actor to enter username and password 4. The system verifies that username and password fields have been filled out and valid 5. The system confirms successful login attempt of the member 6. Use-case ends
Alternate course of action	4.1 If all the fields are not filled out and not matching with the username and/or the password, the system notifies the actor a message as “Invalid Username or Password” and redirect to step 2 of basic course of action to reenter fields.	

Table 3. 1 Use case narrative for User login

Use case Number	UC-02	
Use case Name	Register Member	
Priority	High	
Actor/s	Site Visitor/Applicant, Moderator – Registration & Field Officer	
Description	This use-case describes how an applicant be a member in the Tea Grower Support System	
Precondition	• Visitor surf the TGSS website • Identified the own Tea Land from the system	
Post condition	If the use case was successful, member can obtain TGSS services through the system	
Basic course of action	User Action	System Response
	1. The actor visits online registration page to apply for membership 3. The applicant selects the grower category 5. The applicant enters the details and click on “Apply Now & Continue” button 9. The applicant selects Tea Land Registration No & “Link Tea Land & Finish” button 11. The applicant receives an email which contains registration verification link and verify member registration 13. Moderator (Registration) assign request to Field Officer for Land Inspection 15. Field Officer schedule an Inspection visit to Tea Land for verification 17. Applicant confirms Inspection schedule or suggest optional schedule according to convenience 19. Field Officer inspect tea land, verify documents update the details & confirm registration	2. The system prompts the applicant to select “tea grower category” 4. The system prompts the applicant to fill Member Registration form 6. The system verifies that all mandatory fields have been filled out and all the entries are valid 7. The system checks whether user already exists 8. The system adds applicant details into the database and prompts to select respective Tea Land from the system 10. The system adds details into the database and send email notification to applicants’ email to verify registration 12. The system notifies registration details to Moderator (Registration) for confirmation of registration 14. The system notifies assigned request to Field Officer 16. The system sends email with inspection schedule to Applicant 18. The system notifies confirmed schedule to Field Officer 20. The system sends registration confirmation email with registration details to applicant 21. Use-case ends
Alternate course of action	6.1 If all mandatory fields are not filled out and any of the entry/ies are not valid, the system notifies the actor to fill or enter valid entry/ies and redirect to step 4 of basic course of action. 7.1 If user already exists redirects to basic course of action of step 8 to ends the use-case. 12. 1 If Tea Inspector unable to verify applicant as a tea grower, reject registration request and redirects to basic course of action of step 8 to ends the use-case.	

Table 3. 2 Use case narrative for Member registration

3.4.3 Activity Diagrams

The activity diagrams come under the UML describe dynamic aspects of the system. They are flowcharts that represent the flow from one activity to another. An activity can be described as an operation of the system. One of the activity diagram is shown below as Figure 3.5 and rest were shown in Appendix B.

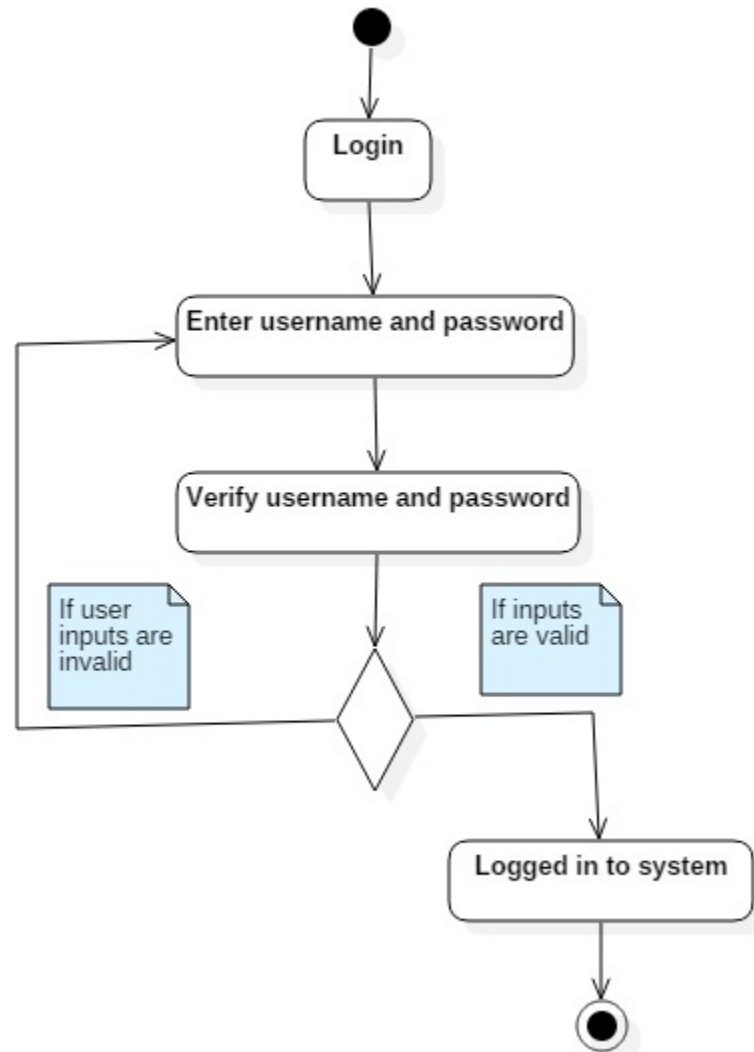


Figure 3. 5 Activity diagram – User login

3.4.4 Entity Relationship Diagrams

The Database design is the structure a database uses to plan, store and manage data. A good database design is important in ensuring consistent data, elimination of data redundancy, efficient execution of queries and high performance application. Taking the time to design a database saves time and frustration during development. In addition, well-designed database ensures ease of access and retrieval of information. The database design for proposed system mainly was done with using entities and attributes

discovered during the analysis. It was used to specify the structure of real world objects found on the problem domain. The following Entity Relationship Diagram (ERD) in Figure 3.6 shows the structure of the Relational Database Model related to the system.

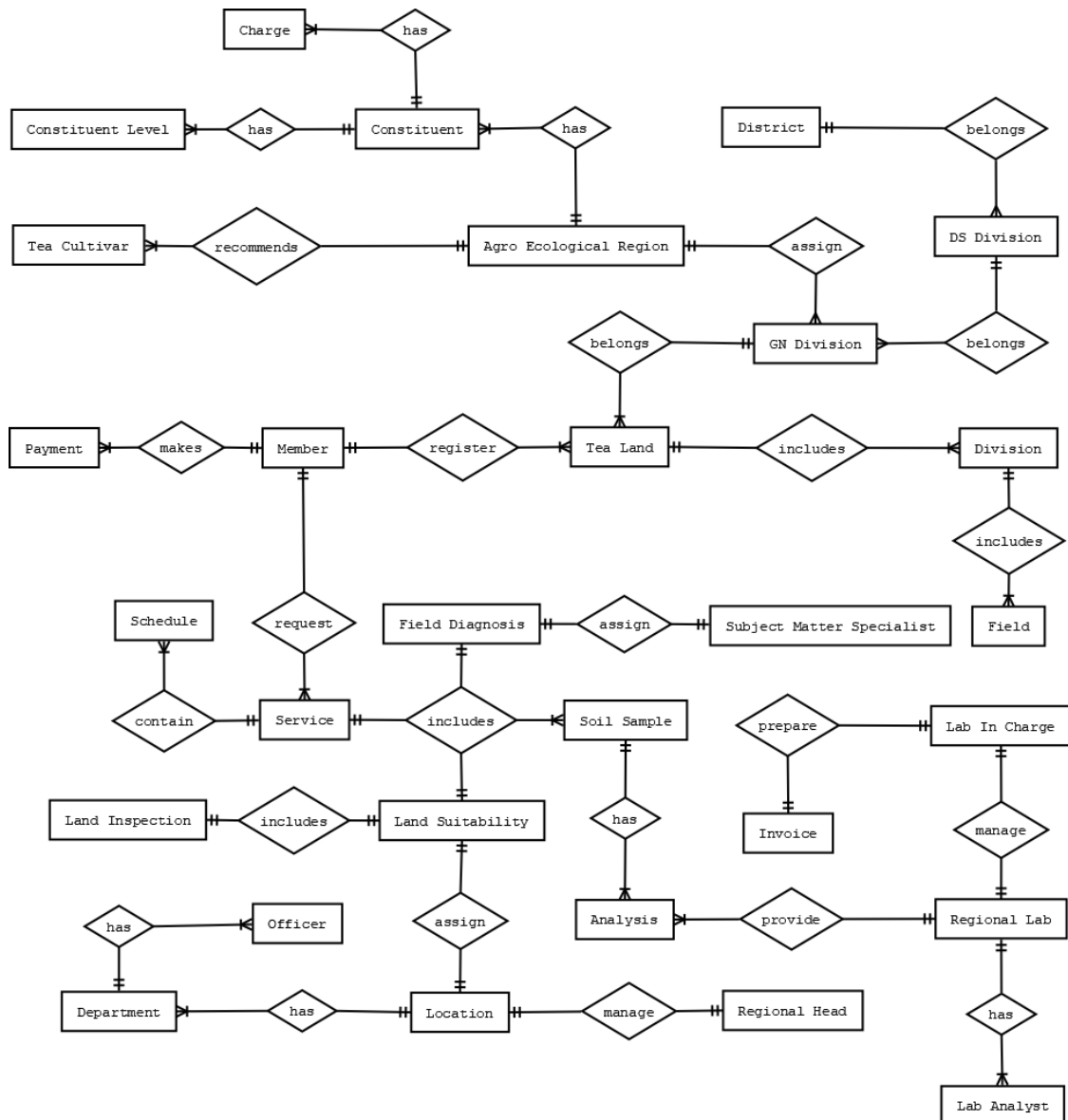


Figure 3. 6 ER Diagram for the proposed TGSS System

In relational database designing, normalization is a systematic way of ensuring that a database structure is suit to protect data and to make it more flexible by eliminating data redundancy and inconsistent dependency. Normalization consists of many forms and done this up to the Third Normal Form (3NF). In the First Normal Form (1NF), eliminate repetition of multi-valued fields, arrange them in a single table and define a primary key to identify each related attribute. Then in the Second Normal Form (2NF) eliminated the partial dependencies, create separate tables and relate tables with foreign keys. Finally,

under the 3NF, it ensured that as much as possible to non-key attributes were set to depend only on primary key.

3.5 User Interface Design

Designing of user attractive interface is one of the important tasks of any system development project. It would be somewhat unique to the client and their requirements. The acceptability and the popularity of a web based system could largely depend on that. Making the user interfaces simple as much as possible and keeping efficiency of process time would be considered thoroughly. Majority of the User Interface (UI) design rules were considered while designing the interfaces as specified below;

- Simple - keep the interface simple
- Consistency - provide clear and consistent design
- Navigation - helps in minimising the number of clicks a user would perform in navigating from one place to another
- User friendliness - use images and icons to make the user feels aware what he/she willing to do
- Simple error handling - display meaningful error messages and appropriate instructions to solve it when user faces with any error
- Look and feel - colours, fonts and their sizes need to be chosen carefully to make it a pleasant environment for scroll up and down since users prefer interfaces are simpler and can be use without learning

The same screen can be built in many different ways, but only a few of them will get the message correctly and result in an easy-to-use software or website. Nailing down a good interface structure is possibly the most important part of designing software. Before any code is written and before the visual design is finalized, hand drawn sketches were shown to the client with the explanation of what it does to eradicate ambiguities and to find out what requirements that have been looking for. It saved lots of time and painful adjustment work later.

This section provides only few main interfaces of the system. The rest of the interfaces can be found in Appendix C.

3.5.1 Website Home Page

The Figure 3.7 shows the home page of the system website that will enable for public. It provides the menu bar and links to navigate easily to user through the website. In

addition, member registration and login links were also included. It contains welcome message on the right, navigation bar with necessary links to other pages on the top right corner and links to navigate member registration and login pages. Also, it facilitates member to check the status of request by entering given reference number and submit to the server. Finally, footer of the page consists of the copyright statement.

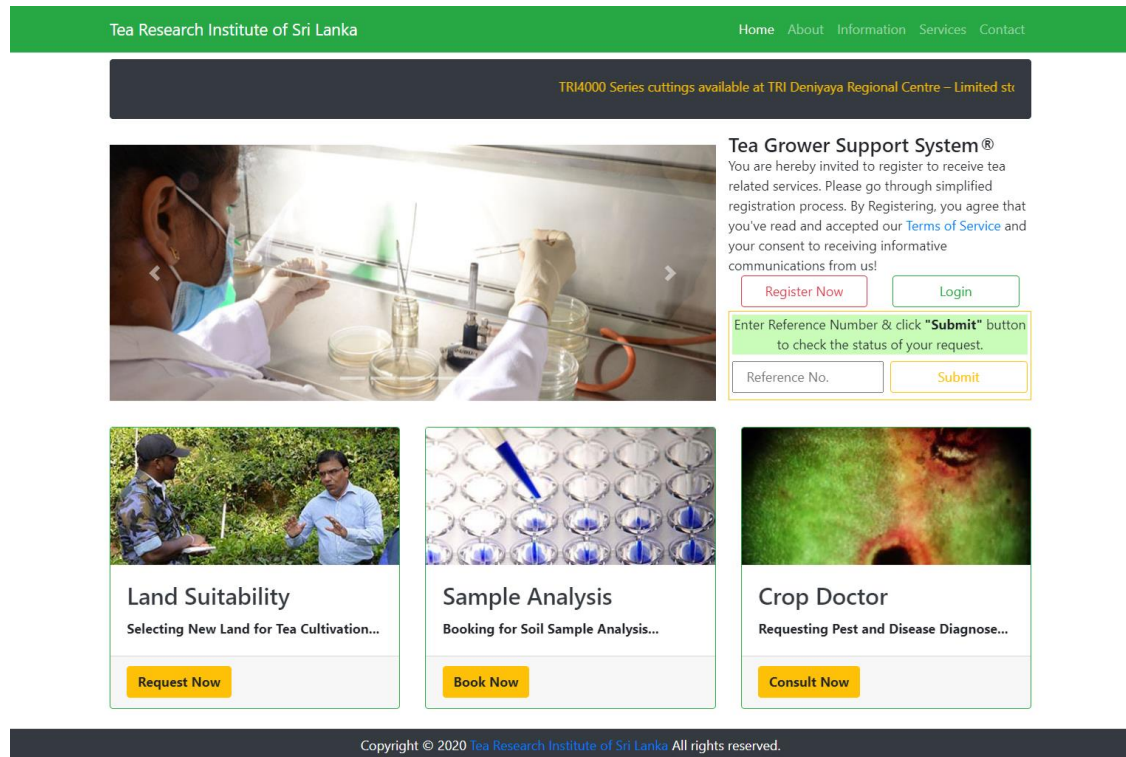


Figure 3. 7 UI of Home Page - TGSS Website

3.5.2 Member Registration

The following interface facilitates tea grower to apply for member registration online, which is shown in Figure 3.8. Targeted users of this interface are tea growers. Therefore, main priority was given to make it simple and self-descriptive while designing.

Tea Grower Support System

Home About Information Services Contact Member

Step 1 of 3: Register to Access e-Services - Personal Details

Read [Terms of Service](#)

Please complete the following fields to register for an account.
Required fields are noted by **bold text and an asterisk (*)**.

Grower Category: * ☐ Corporate ☐ Private

First Name: * **Last Name:** *

Email: * **Password:** *

If you are a corporate user, please enter your official email address provided by the company.

Postal Address: * **Gender:** * ☐ Female ☐ Male

NIC Number: * **Mobile No:** *

☐ I agree to the [Terms of Service](#) *

We're ready to talk about the special needs of you as a tea grower and we're ready to suit our services to meet your needs.
Don't hesitate to contact us, if you have any questions about our services.
We'll answer over the phone or via email.

Hotline: 052 22 58201
Email: support.tgss@tri.lk

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Figure 3. 8 UI of Member Registration (Step 1 of 3) - TGSS Website

3.5.3 User Login

This interface allows users of the client and members to login to the system. The UI is shown in Figure 3.9.

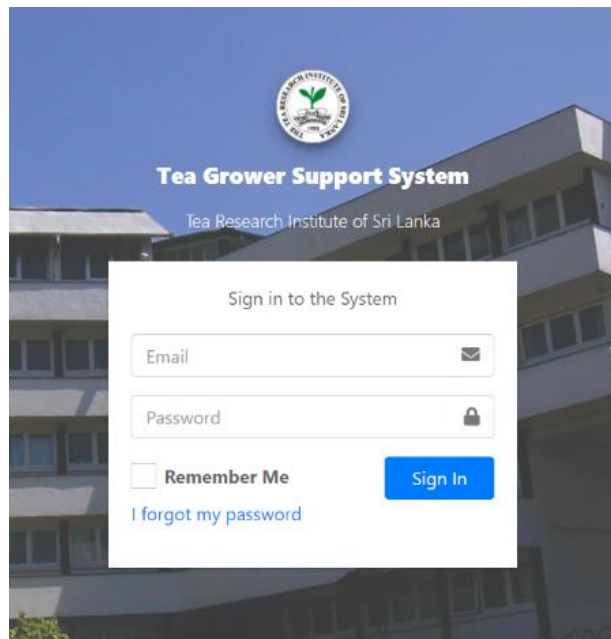


Figure 3. 9 UI of Login Interface

3.5.4 Member Dashboard

Here, Figure 3.10 shows the interface of Member dashboard. It was designed to show the services offered by the client at a glance and quickly make a required service using given shortcut without presenting complex jargon here and there.

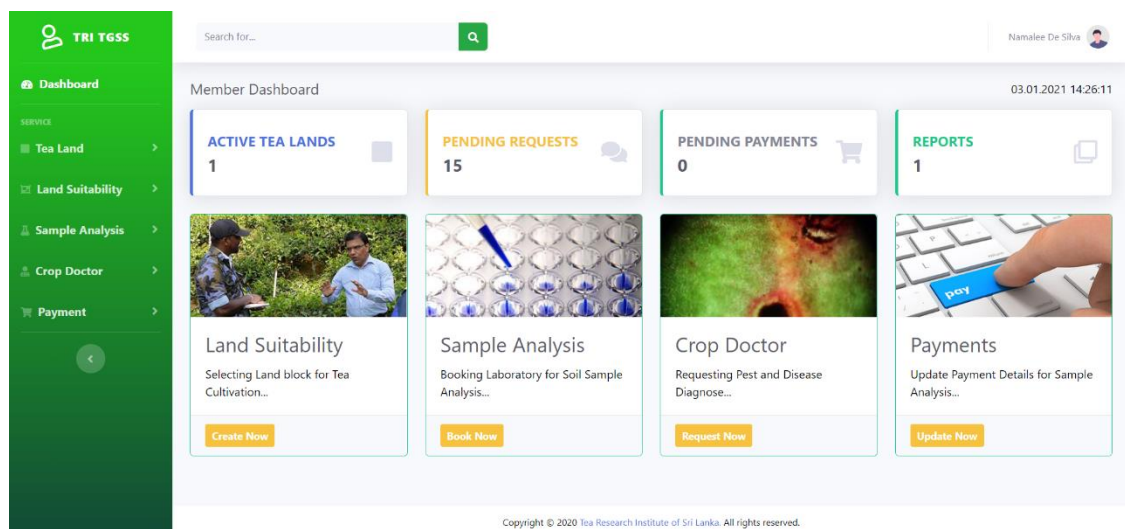


Figure 3. 10 UI of Dashboard - Member

3.5.5 UI of Sample Analysis Request

This UI makes sure that the Member could easily navigate to the service that the TGSS provides. Once the member clicks on “Book Now” button of the “Sample Analysis” card, located in the Member dashboard (depicted in Figure 3.8), following interface will open, which is depicting in Figure 3.11. It makes sure the relevant details regard to generate Sample Analysis request need to be obtained from the member.

TRI TGSS

Search for...

Create Sample Analysis Request

Please provide following details...

The collection, preparation and delivery a sample are important steps in the analytical operation. The samples should be send directly to TRISL Main laboratory or to regional laboratory depending on the region. (* Mandatory Field)

Sample Type: *
☒ Fertilizer
☐ Soil

Tea Land Name: *
-Select Your Tea Land-

Analytical Laboratory: *
Automatically selected

Expected Submission Date: *
mm/dd/yyyy
Do not select weekends & public holidays

Remarks:
example: Due for Pruning

Test Parameters: *

Select testing parameters here.

Constituent	Amount (Rs.)
☐ Phosphorus	540.00
☒ Magnesium	670.00
☒ Potassium	600.00

Total Charge (Rs.): **1270.00**

Calculate Total Charge automatically...

Go Back Reserve Lab Now Reset

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Figure 3. 11 UI of Create Sample Analysis Request

3.5.6 System Dashboard

The primary responsibility of an Administrator/ Admin is to support reliable and effective use of the systems by end users, whether internal employees or tea growers. Therefore, the Admin dashboard was designed to present the data at a glance and easy-to-read format with most relevant information understandable. Here, Figure 3.12 shows the UI of Administrator dashboard.

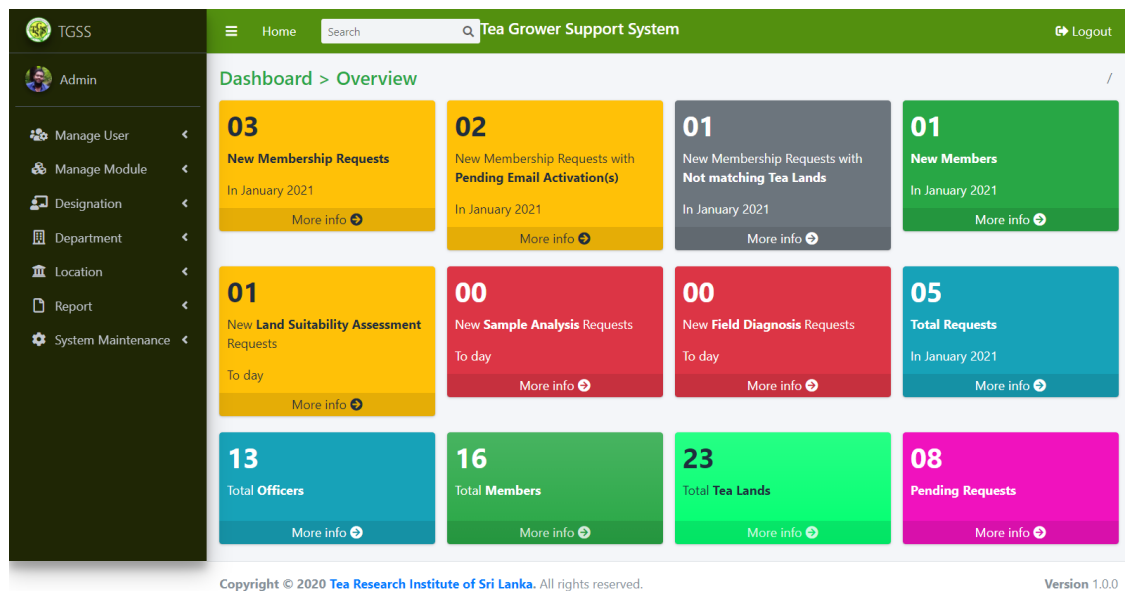


Figure 3. 12 UI of System Dashboard - Administrator

3.5.7 View Users - Dashboard

The UI of View User of the Admin Dashboard shown below as Figure 3.13.

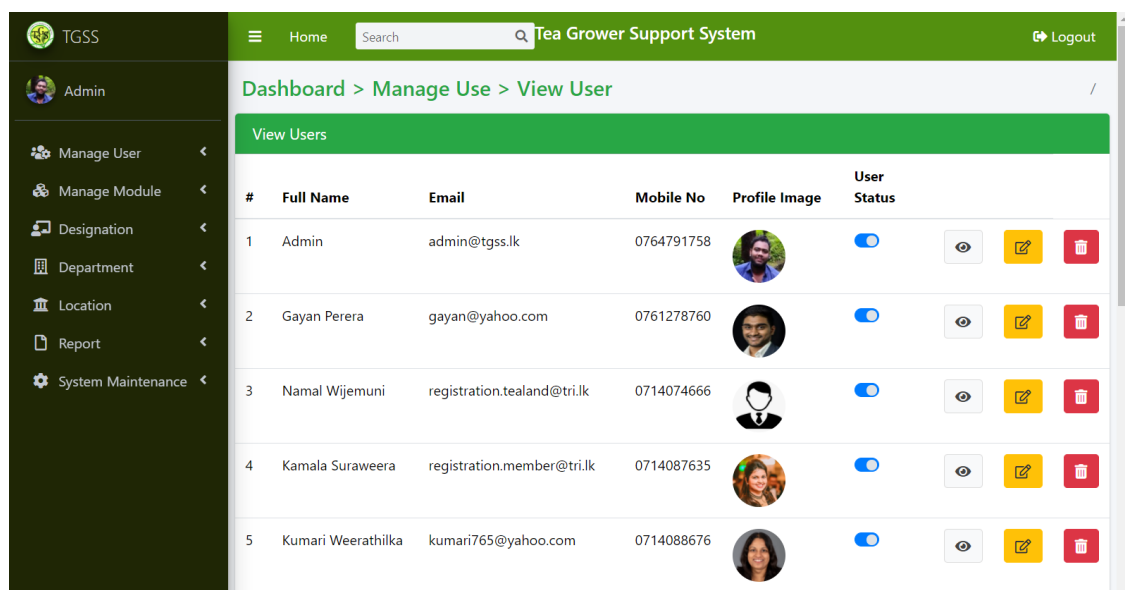


Figure 3. 13 UI of View User – Admin Dashboard

3.5.8 Assign Privilege - Dashboard

The UI of Assign Privilege of the Admin Dashboard shown below as Figure 3.14. Left panel of the interface shows assigned privileges to the users.

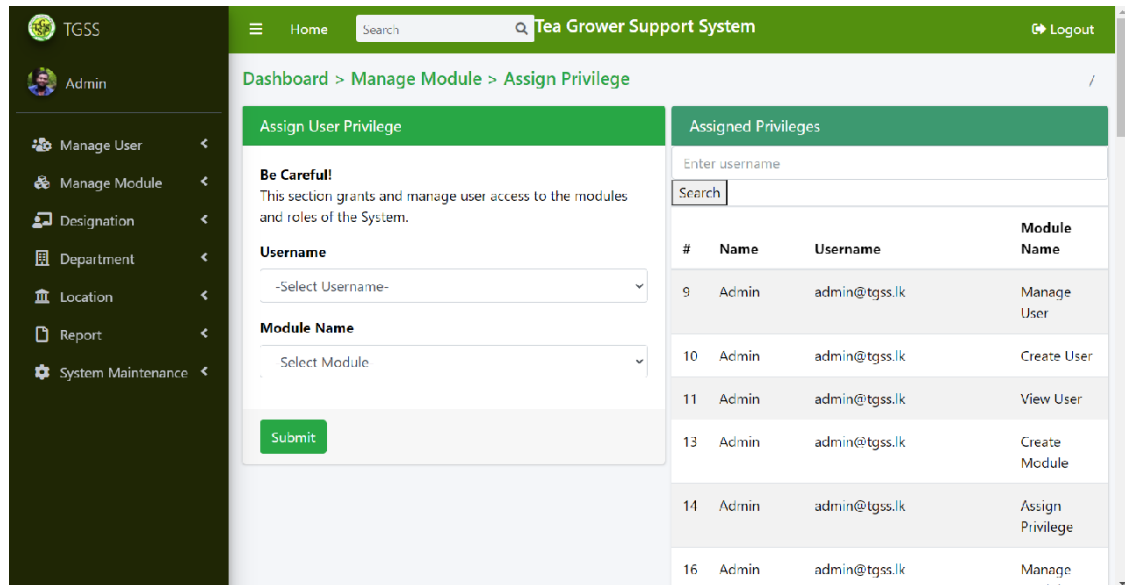


Figure 3. 14 UI of Assign Privilege – Admin Dashboard

3.5.9 New Member Registrations - Dashboard of Moderator

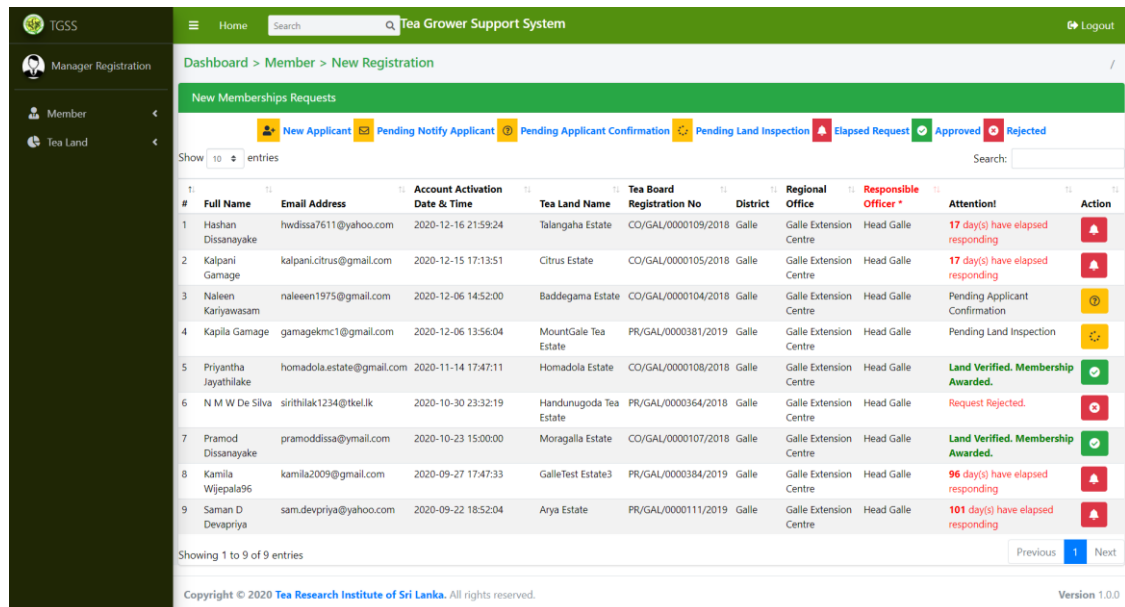


Figure 3. 15 New Member Registrations - Dashboard of Moderator

3.5.10 Request Status - Dashboard of Moderator

Pending Land Inspection

Pending Land Inspection confirmed by Tea Grower

Grower Name: Kalpani Gamage

Category: Corporate

NIC No: 198912345671

Mobile No: 0771264567

Postal Address: Test Address, Baddegama.

Registered On: 2020-12-15 17:10:13

Account Activated On: 2020-12-15 17:13:51

Tentative Finish Date: 2021-01-14

Close

#	Full Name	Email Address
1	Hashan Dissanayake	hwdissa7611@yahoo.com
2	Kalpani Gamage	kalpani.citrus@gmail.com
3	Naleen Kariyawasam	naleen1975@gmail.com
4	Kapila Gamage	gamagekmc1@gmail.com
5	Priyantha Jayathilake	homadola.estate@gmail.com
6	N M W De Silva	sirithilak1234@tel.lk
7	Pramod Dissanayake	pramoddissa@gmail.com
8	Kamila Wijepala96	kamila2009@gmail.com
9	Saman D. Devapriya	sam.devpriya@yahoo.com

District	Regional Office	Responsible Officer *	Attention!	Action
Galle	Galle Extension Centre	Head Galle	17 day(s) have elapsed responding	
Galle	Galle Extension Centre	Head Galle	17 day(s) have elapsed responding	
Galle	Galle Extension Centre	Head Galle	Pending Applicant Confirmation	
Galle	Galle Extension Centre	Head Galle	Pending Land Inspection	
Galle	Galle Extension Centre	Head Galle	Land Verified, Membership Awarded.	
Galle	Galle Extension Centre	Head Galle	Request Rejected.	
Galle	Galle Extension Centre	Head Galle	Land Verified, Membership Awarded.	
Galle	Galle Extension Centre	Head Galle	96 day(s) have elapsed responding	
Galle	Galle Extension Centre	Head Galle	101 day(s) have elapsed responding	

Figure 3. 16 UI for Request Status of New Member Registration

Chapter 4 – Implementation

This chapter describes how the system specification developed by analysis and design phases converting to as an executable system. The TGSS has been implemented by using popular system development tools. They were supported remarkably to the implementation of user-friendly, responsive and satisfied system in timely manner.

4.1 Implementation Environment

The System Implementation Environment is mainly divided into two categories: namely Hardware Environment and Software Environment.

4.1.1 Hardware Environment

Hardware of the development and implementation environments are different. The Table 4.1 described the components that were used in development and implementation phases.

Resource	Development	Implementation
Processor	Intel(R) Core (TM) 8 th Gen i5 1.60 GHz	Intel(R) Core (TM) Quad Core 2.4 GHz
RAM	8 GB	16 GB
Hard Disk Capacity	1 TB	320 GB
Printer	A4 size Inkjet or Laser printer	
Internet	ADSL or 4G connection	

Table 4. 1 Hardware environment

The client's web server is much powerful, compared to the notebook computer used for the development. Since the client expect about 100 unique users will be accessing the site daily, it will be added advantage.

4.1.2 Software Environment

The software and supported technologies that were used in both development and implementation environments are same except operating system and web server used and they were described in the table 4.2. Although the system was developed on Microsoft Windows 10 Pro environment, it was implemented on the Linux platform. When selecting development software, most of them were free and open source (FOSS) which will not cause much trouble when getting the copyrights of the system.

Software	Development	Implementation
Operating System	Microsoft Windows 10 Pro	Ubuntu 20.4 LTS
Web server	XAMPP 8.0.0 Apache 2.4 PHP 8.0 MariaDB 10.4 phpMyAdmin 5.0 XAMPP Control Panel 3.2.4	LAMP stack Apache 2.4 PHP 8.0 MariaDB 10.4 SQLite 2.8 phpMyAdmin 5.0
Database handling	MySQL Workbench 8.0	
Coding	NetBeans IDE 12.0	
Photo editing	Adobe Photoshop CS6	
Design diagrams	StarUML 2.8.1 diaw.exe 0.97.2	
Web browsers	Google Chrome 61.0 Mozilla Firefox 49.0.1 Microsoft Edge 25.10586	
Documentation	MS Word Professional 2016 MS Project 2007 Adobe Acrobat Reader DC	

Table 4. 2 Software environment

4.1.3 Network Environment

Since TGSS is a web based system a dedicated web server and a database server could be used to achieve better performance. However, the system was implemented with available single server. Then the network infrastructure should be configured with a WAN connection, static IP and a web server should be configured with a domain name so that the online users can be accessed to the system through this setup. The system is made accessible to them via internet using a preferred web browser. The staff members can access the system by connecting to the web server locally through their web

browsers. The Figure 4.1 illustrates the Network environment of the actual implementation setup at the clients' premises.

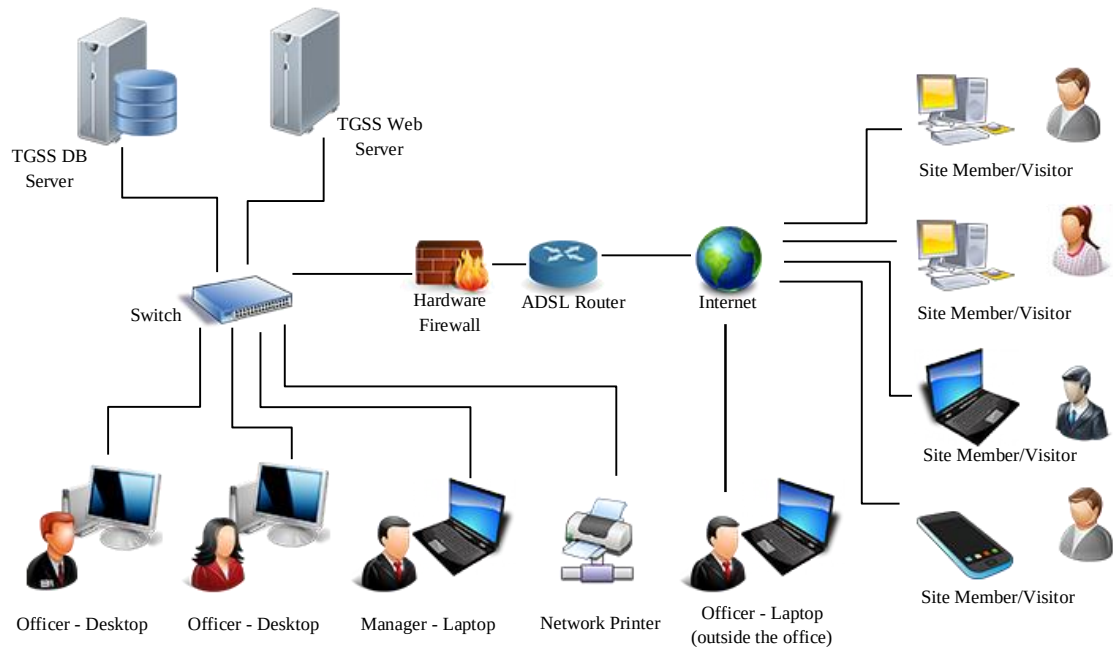


Figure 4. 1 Network environment of the actual implementation

4.2 Development Tools

XAMPP server [21] has been used as it consists Apache as web server, MariaDB as relational database management system (RDBMS) and PHP server-side scripting language for Windows environment. The NetBeans IDE has been used for coding. NetBeans is an open-source integrated development environment (IDE) for programming languages like Java, PHP, C++ etc [19]. It was downloaded from the Apache Friends website [19] since it is available for free under the GNU General Public License (GPL). Bootstrap 4.5 is the newest version, which is the most popular HTML, CSS and JavaScript framework for developing responsive, mobile-first websites [20]. The Ajax (Asynchronous JavaScript and XML) technique was also used to communicate with server to create better, faster and more interactive pages inside the framework.

4.3 Justification for the Implementation Platform

Bootstrap 4.5

It is a free and open-source front-end web framework for designing responsive websites and web applications. The responsive feature means the layout of the web page adjust dynamically, considering the characteristics of the device used whether it is a desktop, laptop, tablet or smart phone. Bootstrap contains HTML- and CSS-based design templates for typography, forms, buttons, navigation and other interface components, as well as optional JavaScript extensions. The JavaScript was used mainly for client-side validations of form data. Unlike many web frameworks, Bootstrap concerns itself with front-end development only. It supports the most of latest browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, Opera, etc., [20]. The Bootstrap toolkit was downloaded from its official website of <http://getbootstrap.com>.

PHP 8.0

The PHP was used as server-side scripting language which was designed for web development. The PHP coding could be used inside the HTML code and Bootstrap framework supports much on that. As an open sourced programming language, PHP has been widely ported and can be deployed on most web servers on almost every operating system and platform, free of charge. The XAMPP Sever itself contains latest PHP version.

MariaDB 10.4

It is an open-source RDBMS and was used to manipulate database. It has a simpler learning curve and require low hardware requirements. MariaDB is a cleaner, simpler system that is easier to learn and easier to manage effectively than many other database systems. It runs as a server providing multi-user access to the database and is also available for free use under the GNU GPL. The XAMPP Sever used for the development environment contains MariaDB itself.

Ajax

Ajax supports updating the system components without refreshing the whole page onl the required field is reloaded. It increases the browser performance and facilitates faster browsing.

Since, TGSS is a web based system, it is platform-independent. It runs on all operating systems with a web browser.

4.4 Module Structure

A module is an individual component which can be used with other components in a system. Identical functions are arranged together with the same programming code and separate functions are created as separate units of code making it available for reuse with other modules. High-level module structure allows to divide a program into smaller sub-programs in order to execute a variety of tasks. Another advantage of this is, it enables developers to work simultaneously and minimizes the time taken for development. A program module is capable of being re-used in a program which minimizes the development of redundant codes. It is also more convenient to reuse a module than to write a program from start. It also requires very little code to be written. Considering the above facts, the TGSS is divided into 8 modules as mentioned in the Section 3.4.1 and described in Table 4.3.

Module	User
User and Role Management	Administrator
Tea Land Registration	Moderator (Registration), Member
Member Registration	Moderator, Field Officer, Member
Land Suitability Assessment	Moderator, Field Officer, Member
Sample Analysis	Moderator, Lab Analyst, Member
Field Diagnosis	Moderator, Subject Matter Expert, Member
Payment	Cashier, Member
Report	Administrator, Moderator, Member

Table 4. 3 Module structure of TGSS

4.5 Major Code Segments

The major code segments of the system have been briefly described accordingly with the functionality/ies and comments.

1. User Login

The code segment given below verifies records of the “tb_user” table and if user name and the password given are matching with user input according to user type (i.e. Office, Member, etc) redirects to relevant user's dashboard.

```

//to start session when user log in to system
session_start();
//this file contain header itself, therefore, no need call header.php
include 'helper.php';
//make database connectivity
include 'conn.php';

if ($_SERVER["REQUEST_METHOD"] == 'POST') {
    //define variables
    $user_name = $password = $user_category = $user_status = $profile_complete = null;
    $e = array();
    //here cleanInput function calling from helper.php
    $user_name = cleanInput($_POST['user_name']);
    $password = cleanInput($_POST['password']);

    if (empty($user_name)) {
        $e['user_name'] = "Username should not be empty....!";
    }
    if (empty($password)) {
        $e['password'] = "Password should not be empty....!";
    }

    if (empty($e)) {
        $sql = "SELECT * FROM tb_user WHERE user_name='$user_name' AND password='". sha1($password) ."'";
        $result = $conn->query($sql);

        if ($result->num_rows == 1) {
            while ($row = $result->fetch_assoc()) {
                $user_category = $row['user_cat_code'];
                $user_status = $row['status'];
                $profile_complete = $row['p_comp_code'];
                $_SESSION['user_id'] = $row['user_id'];
                $_SESSION['profile_image'] = $row['profile_image'];
            }

            if ($user_category=='officer') {
                //officer login
                $sql_officer="SELECT * FROM tb_officer WHERE user_id='".$_SESSION['user_id']."'";
                $result_officer=$conn->query($sql_officer);

                if ($result_officer->num_rows > 0) {
                    while ($row_officer = $result_officer->fetch_assoc()) {
                        $_SESSION['officer_id'] = $row_officer['officer_id'];
                        $_SESSION['first_name'] = $row_officer['first_name'];
                        $_SESSION['last_name'] = $row_officer['last_name'];
                        $_SESSION['login_key'] = 1925;
                    }
                }
                //end officer login
            } elseif ($user_category=='member') {
                //member login
                $sql_member="SELECT * FROM tb_member WHERE user_id='".$_SESSION['user_id']."'";
                $result_member=$conn->query($sql_member);

                if ($result_member->num_rows > 0) {
                    while ($row_member = $result_member->fetch_assoc()) {
                        $member_id = $row_member['member_id'];
                        $_SESSION['member_id'] = $member_id;
                        $_SESSION['first_name'] = $row_member['first_name'];
                        $_SESSION['last_name'] = $row_member['last_name'];
                        $_SESSION['login_key'] = 1975;
                    }
                }
                //end member login
            }
        }
    }
}

```

2. Log out user

If this code segment executes by using “logout.php” it destroys the session created for the specific user and redirects to login page.

```
<?php
//start a session to read already assigned session variable(s)
session_start();
//destroy already created all sessions
session_destroy();
header("Location:login.php");
- ?>
```

3. Validate password strength

When creating user account this validation executes. Here, password length, inclusion of at least one lowercase and uppercase characters are checking. Further, inclusion of at least one number and special character are also checking.

```
// validate password strength
if (empty($e)) {
    $lowercase = preg_match('/[a-z]/', $password);
    $uppercase = preg_match('/[A-Z]/', $password);
    $number = preg_match('/[0-9]/', $password);
    $special_chars = preg_match('/[^\w]/', $password);

    if (strlen($password) < 6) {
        $password_msg .= 'should be at least 6 characters in length, ';
    }
    if (!$lowercase) {
        $password_msg .= 'should include at least one lowercase letter, ';
    }
    if (!$uppercase) {
        $password_msg .= 'should include at least one uppercase letter, ';
    }
    if (!$number) {
        $password_msg .= 'should include at least one number, ';
    }
    if (!$special_chars) {
        $password_msg .= 'should include at least one special character.';
        $pwd_error_count = 5;
    }
    if (!empty($password_msg)) {
        if ($pwd_error_count != 5) {
            $e['password'] = 'Password '. substr($password_msg, 0, -2) . '...!';
        } else {
            $e['password'] = 'Password '. substr($password_msg, 0, -1) . '...!';
        }
    }
}
```


4. Validate image before uploading

This validation also comes in to effect that while creating user account.

```
//validate uploading profile image
if (empty($e) && !empty($_FILES["profile_image"]["name"])) {
    //target folder to upload image file
    $target_dir = "../uploads/profile_image/";
    //obtain name of the selected file and create target folder path + file name
    $target_file = $target_dir . basename($_FILES["profile_image"]["name"]);
    //upload_ok checks whether the upload is ok or not & initialize as 1
    $upload_ok = 1;
    //check image file type by extension using PATHINFO_EXTENSION in lowercase
    $image_file_type = strtolower(pathinfo($target_file, PATHINFO_EXTENSION));

    //check if file is already exists
    if (file_exists($target_file)) {
        $e["profile_image"] = "Sorry, file already exists..!";
        $upload_ok = 0;
    }

    //check if image file is an actual image or fake
    $check_image = getimagesize($_FILES["profile_image"]["tmp_name"]);
    if ($check_image == true) {
        //set profile_image = $check_image["mime"];
        $upload_ok = 1;
    } else {
        $e["profile_image"] = "File is not an image..!";
        $upload_ok = 0;
    }

    //check file size in bytes (0.5MB)
    if ($_FILES["profile_image"]["size"] > 500000) {
        $e["profile_image"] = "Sorry, your file is too large..!";
        $upload_ok = 0;
    }

    //allow selected image file formats only
    if ($image_file_type != "jpg" && $image_file_type != "png" && $image_file_type != "jpeg" && $image_file_type != "gif") {
        $e["profile_image"] = "Sorry, only JPG, JPEG, PNG & GIF files are allowed..!";
        $upload_ok = 0;
    }
}
```

5. Select land type - Create Land Suitability Assessment Request

```
<div class="form-group row">
    <label class="col-form-label col-sm-4 mt-n2">Nature of Land Block:
    <span class="text-danger">*</span></label>
    <div class="col-sm-8">
        <?php
        $sql_landtype = "SELECT * FROM `tb_land_type`";
        $result_landtype = $conn->query($sql_landtype);
        ?>
        <?php
        if ($result_landtype->num_rows > 0) {
            while ($row_landtype = $result_landtype->fetch_assoc()) {
                ?>
                <div class="form-check">
                    <input class="form-check-input" type="radio" name="land_type" id="
                    <?php echo $row_landtype["land_type_code"]; ?>
                    value="<?php echo $row_landtype["land_type_code"]; ?>" <?php if
                    (@$land_type == $row_landtype["land_type_code"]) { ?> checked <?php } ?>
                    onchange="loadAgeOldTeaField(land_type)">
                    <label class="form-check-label" for="<?php echo $row_landtype["land_type_code"]; ?>">
                    <?php echo $row_landtype["land_type"]; ?>
                    </label>
                </div>
            }
        }
        ?>
        <div class="text-danger"><?php echo @$e["land_type"]; ?></div>
    </div>
</div>
```

6. Check last index order of Module available in the system

```
//check existing Index Order values of 2 digit Module IDs (Parent)
if (!empty($module_id)) {
    if (strlen($module_id) == 2) {
        $sql_index_order="SELECT * FROM tb_module WHERE LENGTH(module_id)=2 AND index_order='$index_order'";
        $result_index_order=$conn->query($sql_index_order);
        if ($result_index_order->num_rows>=1) {
            $e['index_order']="This Index Order is taken..!";
        }
        //obtain maximun value of Index Order for Parent Module
        $sql_max2="SELECT MAX(index_order) AS index_order FROM tb_module WHERE LENGTH(module_id)=2";
        $result_max2=$conn->query($sql_max2);
        if ($result_max2->num_rows==1) {
            while ($row_max2=$result_max2->fetch_assoc()) {
                $max=$row_max2['index_order'];
                if ($max>=$index_order){
                    $e['max_id']=$max+1;
                    $e['max_id']=" Suggested Value is ".$e['max_id'];
                }
            }
        }
    }
    //check existing Index Order values of 2 digit Module IDs (Parent)
} elseif (strlen($module_id) == 4) {
    $sql_index_order="SELECT * FROM tb_module WHERE LENGTH(module_id)=4 AND LEFT(module_id,2)="
        . "LEFT('$module_id',2) AND index_order='$index_order'";
    $result_index_order=$conn->query($sql_index_order);
    if ($result_index_order->num_rows>=1) {
        $e['index_order']="This Index Order is taken..!";
    }
    //obtain maximun value of Index Order for Sub Module
    $sql_max4="SELECT MAX(index_order) AS index_order FROM tb_module WHERE LENGTH(module_id)=4 "
        . "AND LEFT(module_id,2)=LEFT('$module_id',2)";
    $result_max4=$conn->query($sql_max4);
    if ($result_max4->num_rows==1) {
        while ($row_max4=$result_max4->fetch_assoc()) {
            $max=$row_max4['index_order'];
            if ($max>=$index_order){
                $e['max_id']=$max+1;
                $e['max_id']=" Suggested Value is ".$e['max_id'];
            }
        }
    }
}
}
}
//end advanced validation
```

7. Cost calculation – Sample Analysis

Following Ajax code pass total charges to show calculated amount according to user preference when member creating Sample Analysis Request.

```
<script>
//ajax code to calculate total charge for analysis
function costCalculation() {
    var total_charge=$("#sample_analysis_request").serialize();
    $.ajax({
        type: 'POST',
        data: total_charge,
        url: 'total_charge.php',
        success: function (data) {
            $("#total_charge").val(data);
        },
        error: function (request, status, error) {
            alert(request.responseText)
        }
    });
}
</script>
```

4.6 Re-used Existing Codes

Aiming to maximize the efficiency and to add more attraction to the system some of the well tested re-usable components were incorporated when implementing. They are illustrated in Table 4.4.

Component	Description
Bootstrap 4.5	HTML- and CSS-based design templates, responsive feature, front-end development, client-side validations are salient features of this framework [20]. Here the AdminLTE Bootstrap Admin Dashboard Template was used to create structure of the user interfaces [23].
CSS re-usable components	Apply to modify color themes of dashboards and styling some parts of the HTML pages
JQuery re-usable components	Implemented using dropdown menus, date & time pickers and accordion layouts
Font Awesome Icons [24]	Freely available set of Font Awesome icons were used to create attractive module structure.
PHPMailer [25]	This function creates email messages from HTML files. This feature is using to communicate in the system.
DataTables [26]	Integration of DataTables facilitates table views in a proper way. Data could be represented tabular way to present to users more attractively.
ChartJS library [27]	This Open Source HTML5 chart generation. This is using for presentation of data in order to take management decisions.
jsPDF AcroForm Plugin [22]	Generate PDF reports at Client Side. This is sing to create transaction reports as well as management reports.
QR Code library [28]	This is an open source PHP library for generating QR codes.
instascan QR Code Scanner [29]	HTML5 QR code scanner is using in the Sample Analysis module of the system to identify labels of samples. There is facility to check the analysis results

	by scanning the QR code which is printed on the invoice.
--	--

Table 4. 4 Re-usable components used to develop the system

It was decided to arrange some identical functions together to implement common programming codes during designing the system. They were made available for reuse with other modules and code segments are given below.

1. Re-usable File Path

```
<!-- Sidebar Menu -->
<nav class="mt-2">
  <ul class="nav nav-pills nav-sidebar flex-column" data-widget="treeview"
    role="menu" data-accordion="false">
    <!-- filter assigned modules & create navigation bar specific to logged user -->
    <!-- obtain parent item/s for menu -->
    <?php $sql_parent="SELECT * FROM `tb_user_module`"
      . " LEFT JOIN tb_module ON tb_module.module_id =
      tb_user_module.module_id "
      . "WHERE tb_user_module.user_id=" . $_SESSION['user_id']
      . " AND LENGTH(tb_module.module_id)=2";
    $result_parent = $conn->query($sql_parent);
    if ($result_parent->num_rows > 0) {
      while ($row_parent = $result_parent->fetch_assoc()) { ?>
    <li class="nav-item has-treeview">
      <a href="#" class="nav-link">
        <i class="nav-icon <?php echo $row_parent['menu_icon']; ?>"></i>
        <!-- create parent module specific to user -->
        <p> <?php echo $row_parent['description']; ?>
        <i class="right fas fa-angle-left"></i>
      </p>
      </a>
      <!-- obtain sub menu item/s to create menu -->
      <?php $sql_sub = "SELECT * FROM `tb_user_module`"
        . "LEFT JOIN tb_module ON tb_module.module_id =
        tb_user_module.module_id "
        . "WHERE tb_user_module.user_id=" . $_SESSION['user_id'] . " "
        . "AND LENGTH(tb_module.module_id)=4 "
        . "AND SUBSTR(tb_module.module_id,1,2) =
        " . $row_parent['module_id'] . " ";
      $result_sub = $conn->query($sql_sub); ?>
      <ul class="nav nav-treeview">
        <?php if($result_sub->num_rows>0) {
          while ($row_sub=$result_sub->fetch_assoc()) {
            //generate path of the page (file)
            $path=$row_sub['module']. "/" . $row_sub['view'] . ".php"; ?>
          <li class="nav-item">
            <a href="<?php echo $root; ?><?php echo $path; ?>" class="nav-link">
              <i class="nav-icon <?php echo $row_sub['menu_icon']; ?>"></i>
              <p><?php echo $row_sub['description']; ?></p>
            </a>
          </li>
          <?php } } ?> </ul> </li> <?php } } ?> </ul>
    </nav> <!-- /.sidebar-menu -->
```

Above code segment creates page access to the user according to assigned modules and access permissions. It dynamically creates menu and sub menu items when user login to the system.

2. Create baseurl

```
<?php
    //creating baseurl
    $root = (!empty($_SERVER['HTTPS'])? 'https' :
'http').'://'.$_SERVER['HTTP_HOST'].'/tgss/';
?>
```

3. Clean input function

It cleans unnecessary data coming as user inputs (i.e. spaces, backslashes, special charaters, etc.)

```
<?php
    //cleaning unnecessary data
    function cleanInput($data=null) {
        $data=trim($data);
        $data= stripslashes($data);
        $data= htmlspecialchars($data);
        return $data;
    }
```

4. Check date has passed

This function checks if user selects previous date or not.

```
    //check date is in the past or not
    function checkDateHasPassed($date=null) {
        $timestamp = new DateTime($date);
        $now = new DateTime();
        if ($timestamp > $now) {
            return false;
        } else {
            return true;
        }
    }
```

Chapter 5 – Evaluation

Software evaluation is a critical element of software quality assurance that represents the review of specification, design and development. The evaluation of performance, from the perspectives of both developers and users, complex systems of hardware and software. Modern computer-based information systems have become increasingly complex because of networking, distributed computing, distributed and heterogeneous databases, and the need to store large quantities of data. When these systems fail, significant breakdowns may arise. A computer system can fail in two major ways. First, functional failure occurs when the system fails to generate the correct results for a set of inputs. Second, performance failure occurs when the system operates correctly but fails to deliver the results in a timely manner. Therefore, in order to prevent these kinds of unnecessary failures, a good system testing has to be done before using it in real environment [30]. The testing is mainly involved with massive set of verification and validation methods. Validation means the system satisfied the requirements and verification refers to the system implements the required functions appropriately.

5.1 Test Plan

The system was tested initially when coding. This code level testing (white box testing) was carried out for to check the correctness of the code. The system functions and behaviors of code modules were also tested for the correctness. An integration testing also done to test integration between related modules (black box testing). In summary, TGSS was tested according to the preplanned test cases and their results are illustrated accordingly.

5.2 Test Cases

The tables 5.1 illustrate few of the test cases which are used for testing. The rest of them can be found in Appendix E.

ID	Test Description	Testing Procedure	Expected Results	Priority
General Evaluation – Test Plan				
1	Page Title	Appropriateness – Main site	Appropriate	High
2	Font size	Appropriateness – Main site	Appropriate	High
3	Font type	Appropriateness – Main site	Appropriate	Medium
4	Spelling of page content	Check for spelling mistakes - Main site	No spelling errors	High
5	Grammar of page content	Check for grammar mistakes - Main site	No grammar errors	High
6	Page navigation	Click on links/ buttons - Main site	Redirects to relevant page	High
7	Relevancy of images	Appropriateness – Main site	Appropriate	High
9	Responsiveness of the interfaces	Resizing interface - 19-inch wide screen monitor	Appropriate	High
10		Resizing interface - 17-inch standard square shape monitor	Appropriate	High
11		Resizing interface - 15.6-inch wide laptop screen	Appropriate	High
12		Resizing interface - 8-inch tablet screen	Appropriate	High
13		Resizing interface - 5.5-inch smartphone screen	Appropriate	High
Common Functions – Test Plan				
14	Login in to the system	Submit login with empty field	An error message will appear	High
15	Login in to the system (Negative)	Enter an invalid user name or password	An error message/s will appear	High
16	Login in to the system (Positive)	Enter valid user name and password	Successfully logged in to the system and redirects to dashboard	High

17	Display user name after login	Logged in to the system with proper authentication	Display first name and last name on the top status bar	Medium
18	Display profile picture after login	Logged in to the system with proper authentication	Display profile picture on the top status bar	Medium
19	Privileged module access according to user type	Logged in to the system with proper authentication	Display assigned modules on the left navigation pane of the dashboard	High
20	Logout form the system	Click on the logout button	The user logged out from the system and redirect to the login page	High
User and Module Management – Test Plan				
21	Register an officer	Fill all the * marked inputs parameters of the officer who is not in the system	Register the officer and display success message	High
22	Validate empty filed of register officer form	User hits submits form with all mandatory fields empty	Display error messages	High
23	Check the given phone no has valid length	Enter phone with invalid length	Display error message	High
24	Invalid first name given	Enter invalid characters	Display error message	High
25	Invalid email given as user name	Validate email address is in correct format	Display error message	High
26	Insert too large capacity file as profile image	Validate file size of the selected profile image in allowed capacity	Display error message	High
27	Upload different type of file as profile image	Validate if the image file is an actual image or fake	Display error message	High

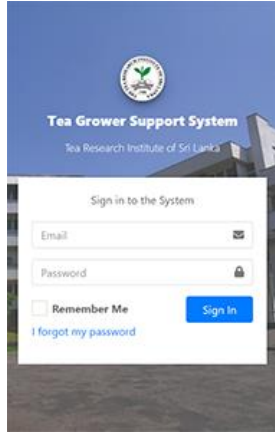
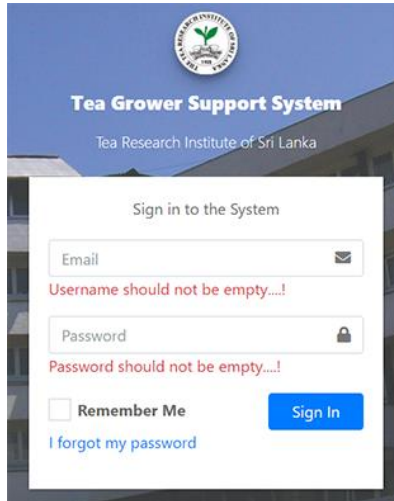
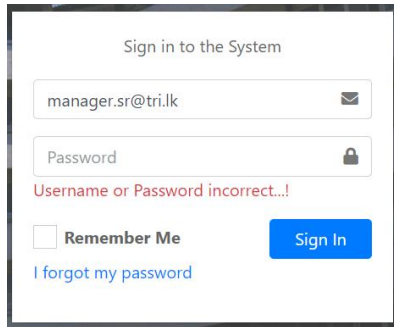
28	Enter username	Check whether username already taken or not in the database	Display error message	High
29	View profile of officer already in the system	Open View user menu and click on View icon of the selected officer	Display profile of the officer	High
30	Modify details of registered officer	Edit relevant field(s) and click Update button	Update the database with new details	High
31	Delete registered officer profile	Select relevant officer record and click 'Delete' icon	Prompt confirmation window, if user selects 'Yes' delete the officer	High
32	Assign new role to user	Select user name, module name then click submit button	Role assigned to user account and display success message	High
33	Assign already assigned role to user	Select user name, module name then click submit button	Display error message	High
Member Registration - Test Plan				
34	Applicant try to select already assigned tea Land	Applicant selects already linked Tea Land	Display error message	High

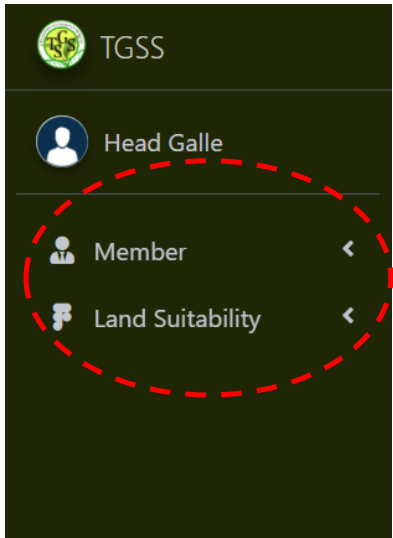
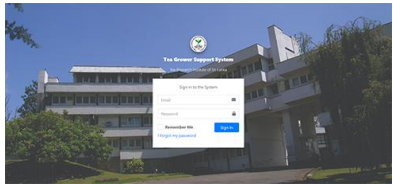
Table 5. 1 Test Plan for the system

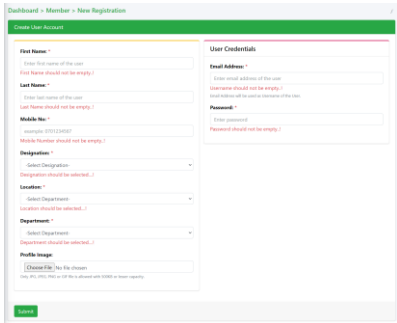
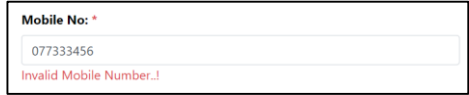
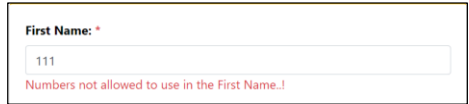
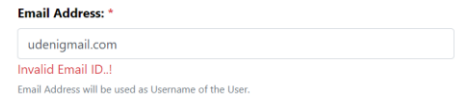
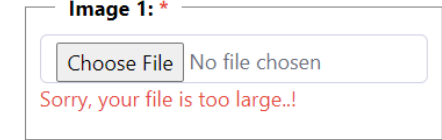
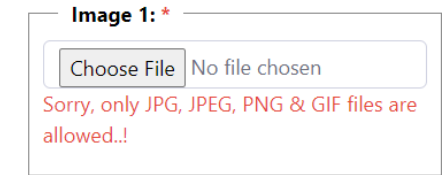
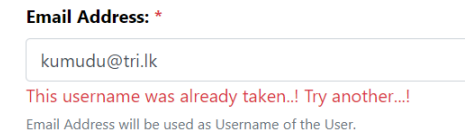
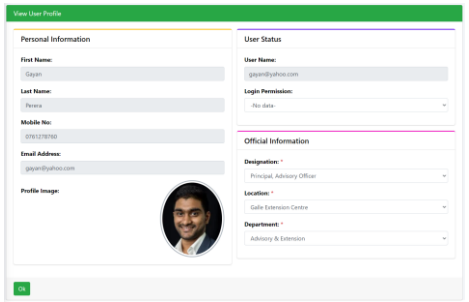
5.3 Test Data and Results

The data required to test the system functions, were obtained from document samples which were gathered during analysis. All data inserting fields were tested with sample data. Most of the test results were successful and few number of error were identified. Error corrections were also done after performing testing. The test results are given below in Table 5.2 and comprehensive test data and test results were included in Appendix E.

ID	Test Description	Testing Procedure	Actual Output	Status
General Evaluation – Test Plan				
1	Page Title	Appropriateness – Main site	Appropriate	Pass
2	Font size	Appropriateness – Main site	Appropriate	Pass
3	Font type	Appropriateness – Main site	Appropriate	Pass
4	Spelling of page content	Check for spelling mistakes - Main site	No spelling errors	Pass
5	Grammar of page content	Check for grammar mistakes - Main site	No grammar errors	Pass
6	Page navigation	Click on links/ buttons - Main site	Redirects to relevant page	Pass
7	Relevancy of images	Appropriateness – Main site	Appropriate	Pass
9	Responsiveness of the interfaces	Resizing interface - 19-inch wide screen monitor	Appropriate	Pass
10		Resizing interface - 17-inch standard square shape monitor	Appropriate	Pass

11		Resizing interface - 15.6-inch wide laptop screen	Appropriate	Pass
12		Resizing interface - 8-inch tablet screen	Appropriate	Pass
13		Resizing interface - 5.5-inch smartphone screen		Pass
Common Functions – Test Plan				
14	Login in to the system	Submit login with empty field		Pass
15	Login in to the system (Negative)	Enter an invalid user name or password		Pass

16	Login in to the system (Positive)	Enter valid user name and password		Pass
17	Display user name after login	Logged in to the system with proper authentication		Pass
18	Display profile picture after login	Logged in to the system with proper authentication		Pass
19	Privileged module access according to user type	Logged in to the system with proper authentication		Pass
20	Logout from the system	Click on the logout button		Pass
User and Module Management – Test Plan				
21	Register an officer	Fill all the * marked inputs parameters of the officer who is not in the system	Register the officer and display success message	Pass

22	Validate empty filed of register officer form	User hits submits form with all mandatory fields empty		Pass
23	Check the given phone no has valid length	Enter phone number with invalid length		Pass
24	Invalid first name given	Enter invalid characters		Pass
25	Invalid email given as user name	Validate email address is in correct format		Pass
26	Insert too large capacity file as profile image	Validate file size of the selected profile image in allowed capacity		Pass
27	Upload different type of file as profile image	Validate if the image file is an actual image or fake		Pass
28	Enter username	Check whether username already taken or not in the database		Pass
29	View profile of officer already in the system	Open View user menu and click on View icon of the selected officer		Pass

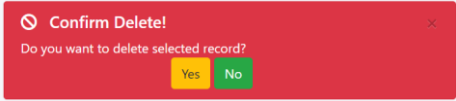
30	Modify details of registered officer	Edit relevant field(s) and click Update button	Update the database with new details	Pass
31	Delete registered officer profile	Select relevant officer record and click 'Delete' icon		Pass
32	Assign new role to user	Select user name, module name then click submit button	Role assigned to user account and display success message	Pass
33	Assign already assigned role to user	Select user name, module name then click submit button	Display error message	Pass

Table 5. 2 Test Results

5.3 User Acceptance

After hosting on the local server at client's site, the TGSS was allowed to work to selected set of users of the client. They were asked to report any errors and malfunctions that they could experience. The selected user activities were also monitored. Subsequently, some issues were reported and corrected time to time. Chosen set of tea growers were also invited use the system and obtained their feedback also. Some of the suggestions requested by the client were implemented too. Due to that client is in a view that that the project performance was partially satisfied and met the requirement.

Chapter 6 – Conclusion

Being the second oldest, tea research institute in the world, in this information age, management of the Tea Research Institute of Sri Lanka came across to very important decision to have this kind of system. Since, the client generates and disseminates scientific knowledge and technologies necessary for improving and sustaining tea cultivation and processing, we were found excellent solution for the problem of inefficiency of information channels, using online information service portal which link the client and tea growers.

Though, legacy communication channels are proved to be time consuming and poor in reaching large number of tea growers within shorter time effectively. But the tea growers would certainly require accessing new technologies developed by TRI and updated information to facilitate their decision making.

The implementation of TGSS has quite an impact on dissemination of scientific knowledge to the tea growers. This will be a versatile system for which tea land data can be entered and online by tea growers. Further, email alerts are used for fast communication. Moreover, the system will generate reports for management to facilitate decision making. According to the ease of use and efficient communication, it will enhance the relationship between the tea grower and TRI.

6.1 Problems Encountered

One of the major problem faced during the development phase was lack of thorough knowledge regarding the development framework and tools. Online materials such as forums, tutorials, presentations and information websites were used to acquire more knowledge.

Some of the detailed requirements were unable to gather due to pandemic restrictions prevailed in the country during early stages. Therefore, code development was unable to complete as scheduled.

Other major issue raised was after the design was finalized the client require to add some features those were not initially agreed and also time limits. Therefore, it was not agreed to accommodate such changes.

6.2 Lessons Learnt

The most important lesson was to aim the deadline and work. In this case time management and prioritizing the work very much help in software development life cycle. The knowledge acquired by starting the project from feasibility study to handing over to the system educate me vastly. In addition to programming, report writing and interpersonal skills were also improved.

6.3 Future Improvements

Though the Tea Grower Support System has fulfilled requirements requested by the client, some additional features demanded may be done in future.

1. Develop Mobile App integrated with TGSS since tea growers mostly at field and thereby it will be more helpful getting relevant data from the ground
2. Integrate payment gateway to provide online payment and ordering facility their product such as research articles and other educational publications, analytical report charges, etc.
3. Integrate SMS gate way to communicate request status and new information to the tea grower.
4. Report previewing of analytical test such as soil, fertilize, nematode, leaf, moisture, etc
5. Online forum could be implemented to discuss the technical matters regarding tea cultivation.

References

- [1] 2016 Annual Report, Tea Research Institute of Sri Lanka, Talawakelle, 4, 2016.
- [2] Tea Research Institute of Sri Lanka, 2020. [Online]. Available: <http://www.tri.lk> [Accessed: 06 Jun, 2020].
- [3] Wikipedia.org - Functional requirement, 2020. [Online]. Available: https://en.wikipedia.org/wiki/Functional_requirement. [Accessed: 10 Jun, 2020].
- [4] Tea Control Act | Volume VIII, Laws of Sri Lanka, Blackhall Publishing, Ireland, 2018. [Online] Available: <http://www.srilankalaw.lk/Volume-VIII/tea-control-act.html> [Accessed: 10 Mar, 2020].
- [5] TRISL Advisory Circular: Guidelines on Land Suitability Classification for Tea, Tea Research Institute of Sri Lanka, Talawakelle, No. LU1 (Serial No. 16/03), 2003.
- [6] Natural Resources Management Center, Department of Agriculture, Sri Lanka, 2020. [Online]. Available: <http://nrmc.lk/NRMC/> [Accessed: 17 May, 2020].
- [7] Hand Book on Tea Tea Research Institute of Sri Lanka, Talawakelle, 12 2008.
- [8] TRISL Advisory Circular: The Suitability of Tea Clones for the Different Regions, Tea Research Institute of Sri Lanka, Talawakelle, No. PN1 (Serial No. 6/02), 2002.
- [9] Non-functional Requirements: Examples, Types, How to Approach, 2019. [Online]. Available: <https://www.altexsoft.com> [Accessed: 02 Jul, 2020].
- [10] Farmers' Portal, Department of Agriculture & Cooperation and Farmers Welfare, Ministry of Agriculture and farmers Welfare, Government of India, 2020. [Online]. Available: <https://farmer.gov.in/FarmerHome.aspx> [Accessed: 20 Jul, 2020].
- [11] Soil Test Pro, TapLogic LLC, 2020. [Online]. Available: <https://www.soiltestpro.com/>. [Accessed: 02 Jul, 2020].
- [12] Plantix, Progressive Environmental & Agriculture Technologies, Germany, 2020. [Online]. Available: <https://plantix.net/en/>. [Accessed: 04 Aug, 2020].
- [13] CropDiagnosis, Ergobyte Informatics S.A., Greece, 2020. [Online]. Available: <https://www.cropdiagnosis.com/portal/crops/en/home#about> [Accessed: 04 Aug, 2020].
- [14] T.A. Alspaugh, Software Process Models, 2019. [Online]. Available: <https://www.thomasalspaugh.org/pub/fnd/softwareProcess.html>. [Accessed: 20 Jul, 2020].
- [15] M. A. Awad, "A Comparison between Agile and Traditional Software Development

Methodologies,” School Comput. Sci. and Software Eng., Univ. Western Australia, 2005.

[16] K. Suguna and S. Babu, “Comparing traditional and agile software development methodology using mobile application,” *Int. J. Comput. Sci.*, vol. 5, pp. 40-46, Nov. 2017.

[17] UML - Quick Guide – Tutorialspoint, 2020. [Online]. Available: https://www.tutorialspoint.com/uml/uml_quick_guide.htm. [Accessed: 09 Jul, 2020].

[18] Magic Web Solutions - The benefits of web-based applications, 2020. [Online]. Available: <http://www.magicwebsolutions.co.uk/blog/the-benefits-of-web-based-applications.htm>. [Accessed: 10 Jul, 2020].

[19] Apache NetBeans, 2020. [Online]. Available <https://netbeans.org>. [Accessed: 06 Mar, 2020].

[20] Bootstrap Get Started, 2020. [Online]. Available: <https://getbootstrap.com/>. [Accessed: 06 Mar, 2020].

[21] XAMPP Installers and Downloads for Apache Friends, 2020. Available: <https://www.apachefriends.org>. [Accessed: 06 Mar, 2020].

[22] jsPDF AcroForm Plugin, 2016. [Online]. Available: <https://cdnjs.cloudflare.com/ajax/libs/jspdf/1.3.2/jspdf.min.js>. [Accessed: 06 Jan, 2020].

[23] AdminLTE, 2019. [Online]. Available: <https://adminlte.io>. [Accessed: 20 May, 2020].

[24] Font Awesome Icons, 2020. [Online]. Available: <https://fontawesome.com>. [Accessed: 22 Jun, 2020].

[25] PHPMailer. [Online]. Available: <https://github.com/PHPMailer>. [Accessed: 07 Jan, 2020].

[26] DataTables, 2021. [Online]. Available: <https://datatables.net>. [Accessed: 21 May, 2020].

[27] Chart.js. [Online]. Available: <https://www.chartjs.org>. [Accessed: 06 Jan, 2020].

[28] PHP QR Code encoder, 2010. [Online]. Available: <http://phpqrcode.sourceforge.net>. [Accessed: 18 Dec, 2020].

[29] Instascan QR Code Scanner, 2016. [Online]. Available: <https://github.com/schmich/instascan>. [Accessed: 21 Nov, 2020].

[30] Computer-system evaluation | Article about Computer-system evaluation by The Free Dictionary. [Online]. Available: <http://encyclopedia2.thefreedictionary.com/Computer-system+evaluation> [Accessed: Oct 09, 2017].

Appendix A - System Manual

This section provides information on how to setup the system and necessary basic configurations is also given. It enables anyone who wishes to continue or use this project work and perform future improvements.

Hardware Requirements

Resource	Requirement
Processor	Intel(R) Core (TM) i5 2.3 GHz or later
RAM	4 GB or higher
Hard Disk Capacity (for the system)	1 GB or higher
Hard Disk Capacity (to upload files)	Unlimited
Display unit	At least 1024 x 768 dpi resolution
CD-ROM Drive	Required
Internet connectivity	Required

Software Requirements

Software	Development	Implementation
Operating System	Microsoft Windows 10 Pro	Ubuntu 20.4 LTS
Web Server	XAMPP 8.0.0 Apache 2.4 PHP 8.0 MariaDB 10.4 phpMyAdmin 5.0 XAMPP Control Panel 3.2.4	LAMP stack Apache 2.4 PHP 8.0 MariaDB 10.4 SQLite 2.8 phpMyAdmin 5.0
Web Browsers	Google Chrome 61.0 Mozilla Firefox 49.0.1 Microsoft Edge 25.10586	

	Microsoft Internet Explorer 11.0.24	
--	--	--

Appendix B - Design Documentation

Use Case Narratives of the System

1. Registration of Tea Land

Use Case	Tea Land Registration
Actor(s)	Administrator
Description	Manage details of all the tea lands in Sri Lanka
Pre-conditions	Obtain latest Register of Tea Estates from the Tea Commissioner of Sri Lanka
Flow of Events	1. The Administrator checks the system for the existence of the record of selected tea land 2. If not, fill the relevant information of tea land 3. Submit to update the DB of the system
Post conditions	Tea land record is updated

2. Registration of Member

Use Case	Member Registration
Actor(s)	Site Visitor
Description	Facilitate to get membership of the system to a tea grower
Pre-conditions	Visitor should be a tea grower in Sri Lanka
Flow of Events	1. Tea grower visits of Home Page of the system/website. 2. Fill relevant information including valid email address by the tea grower in the signup form of Home Page 3. Tea grower submits the information and update the DB 4. Send an email with a confirmation link to the visitor's email address
Post Conditions	Membership record of Tea Grower is created in the DB

3. Validate New Member by Email

Use Case	Validate new member by email
Actor(s)	Site Visitor, Email Gateway
Description	Validate a site visitor by an email when registering to the system
Pre-conditions	Registration details with valid email address should be submitted by the site visitor
Flow of Events	1. Receive the email with validation link 2. Site visitor clicked on the confirmation link which is embedded in the email 3. Visitor's record of the DB will be updated
Post conditions	Site visitor will be considered as a member of the system

4. Mange Field Data

Use Case	Manage Field Data
Actor(s)	Member
Description	Facilitate to manage individual fields of respective tea land
Pre-conditions	1. Authority will only be available for a member registered under relevant tea land 2. Respective division should be available in the DB, since fields of a tea land is coming under a division
Flow of Events	1. List all the divisions belongs to respective tea land 2. Sub Flow: Adding a field a. If relevant division is available, fill details of respective filed b. Submit to the system and update DB 3. Sub Flow: Edit field data a. If the selected field is available, edit relevant data b. Update the DB 4. Sub Flow: Delete a field a. If the selected field does not have yield data, delete respective filed from the DB
Post conditions	Relevant filed data will be updated in the DB

5. Photograph Submission

Use Case	Submit Photograph
Actor(s)	Member, Email Gateway
Description	Facilitate member to submit tea filed/plant photographs which need to diagnose whether infected with a disease or damage done due to a pest attack
Pre-conditions	Need to be a member of the system and logged in
Flow of Events	1. Fill necessary details regarding the location i.e. Name of the Division, Name of the Field 2. Upload photographs 3. User submits request to the system 4. Send email notification to the Administrator
Post conditions	DB will be updated with given information

6. Prepare Fertiliser Estimation Report

Use Case	Prepare Fertiliser Estimation Report
Actor(s)	Content Manager, Member
Description	Facilitate to calculate fertiliser requirement and management of fertiliser applications for requested tea land by individual tea field
Pre-conditions	1. DB need to be updated with all relevant data of tea land including divisions, fields, yield data of two or more consecutive pruning cycles 2. A request should be made by a member to prepare
Flow of Events	1. Check for all required up to date data will be available in the DB 2. Calculate highest potential yield of respective pruning cycle year for each and every field

	3. Select Nitrogen (N) requirement corresponding to potential yield for selected fields separately 4. Choose suitable basal fertiliser mixture for individual tea fields according to obtained N requirement, field type and region located in the tea land 5. Calculate required quantities of chosen fertiliser mixture and urea for fields separately 6. Calculate the quantity per application and number of applications that is practically possible to apply for selected field and need to it for all the fields and stored in the DB
Post conditions	DB will be updated with formulated estimation details

Appendix C - User Documentation

This supplementary is for the use of system users. This system consists five types of user roles and privileges increasing in the order mentioned.

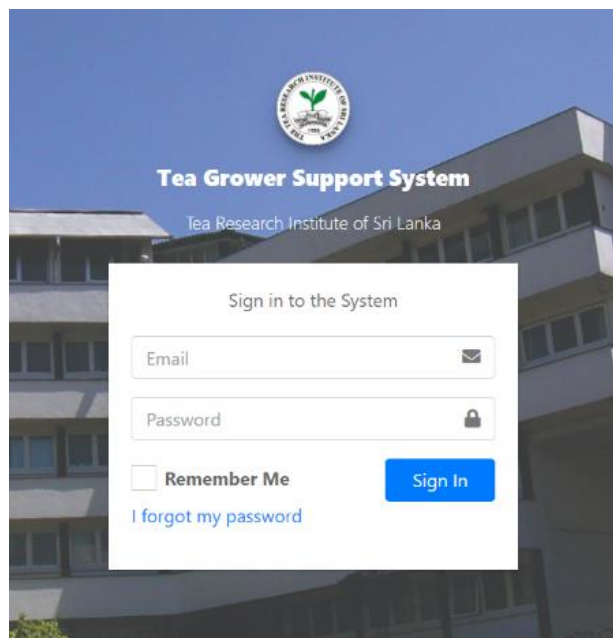
1. Visitors
2. Members
3. Administrator
4. Manager
5. User

The administrator has super power to access all the modules and functionalities in the system. The system consists of two major sections as Main website and backend office application.

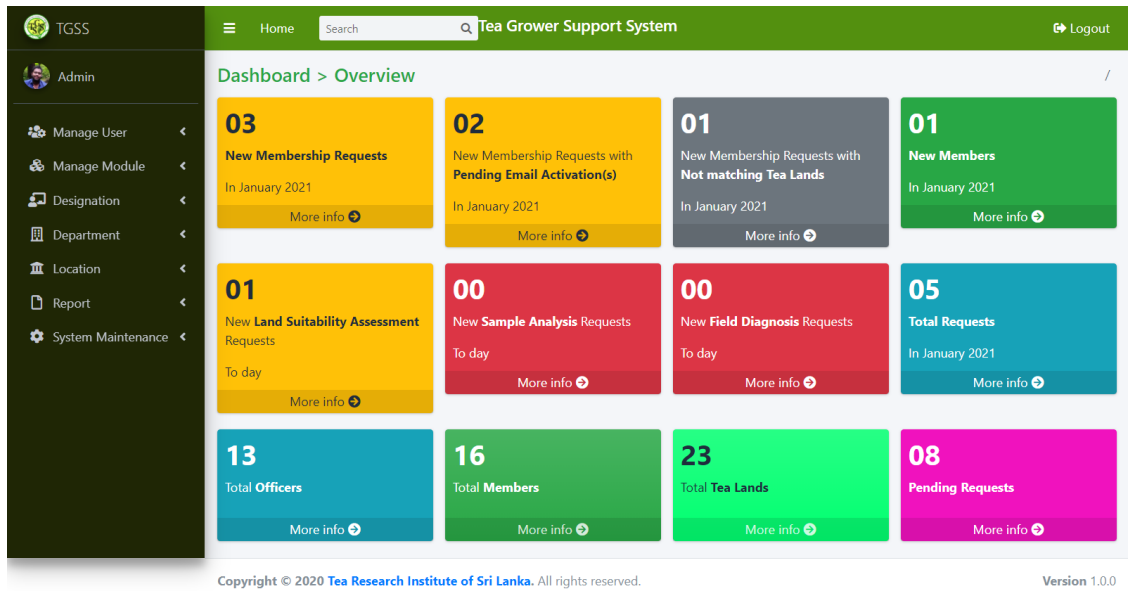
Section 1

User: Officer

Login Interface



Administrator Dashboard

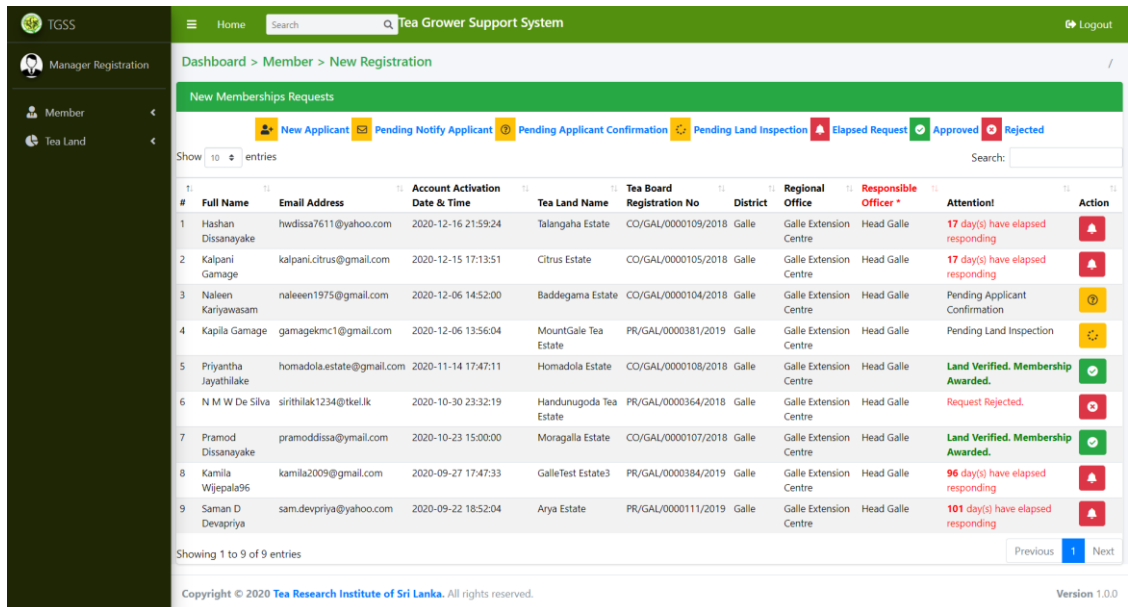


The Administrator Dashboard (TGSS) features a sidebar with navigation options: Admin, Manage User, Manage Module, Designation, Department, Location, Report, and System Maintenance. The main content area displays a grid of 12 summary cards for various metrics:

- 03** New Membership Requests (In January 2021)
- 02** New Membership Requests with Pending Email Activation(s) (In January 2021)
- 01** New Membership Requests with Not matching Tea Lands (In January 2021)
- 01** New Members (In January 2021)
- 01** New Land Suitability Assessment Requests (To day)
- 00** New Sample Analysis Requests (To day)
- 00** New Field Diagnosis Requests (To day)
- 05** Total Requests (In January 2021)
- 13** Total Officers
- 16** Total Members
- 23** Total Tea Lands
- 08** Pending Requests

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Manager Dashboard



The Manager Dashboard (TGSS) shows the 'New Registration' page with a table of new membership requests. The table includes columns for #, Full Name, Email Address, Account Activation Date & Time, Tea Land Name, Tea Board Registration No, District, Regional Office, Responsible Officer, Attention!, and Action.

#	Full Name	Email Address	Account Activation Date & Time	Tea Land Name	Tea Board Registration No	District	Regional Office	Responsible Officer *	Attention!	Action
1	Hashan Dissanayake	hwdissa7611@yahoo.com	2020-12-16 21:59:24	Talangaha Estate	CO/GAL/0000109/2018	Galle	Galle Extension Centre	Head Galle	17 day(s) have elapsed responding	
2	Kalpani Gamage	kalpani.citrus@gmail.com	2020-12-15 17:13:51	Citrus Estate	CO/GAL/0000105/2018	Galle	Galle Extension Centre	Head Galle	17 day(s) have elapsed responding	
3	Naleen Kariyawasam	naleen1975@gmail.com	2020-12-06 14:52:00	Baddegama Estate	CO/GAL/0000104/2018	Galle	Galle Extension Centre	Head Galle	Pending Applicant Confirmation	
4	Kapila Gamage	gamakekmc1@gmail.com	2020-12-06 13:56:04	MountGale Tea Estate	PR/GAL/0000381/2019	Galle	Galle Extension Centre	Head Galle	Pending Land Inspection	
5	Priyantha Jayathilake	homadola.estate@gmail.com	2020-11-14 17:47:11	Homadola Estate	CO/GAL/0000108/2018	Galle	Galle Extension Centre	Head Galle	Land Verified. Membership Awarded.	
6	N M W De Silva	sirithilak1234@tkei.lk	2020-10-30 23:32:19	Handunugoda Tea Estate	PR/GAL/0000364/2018	Galle	Galle Extension Centre	Head Galle	Request Rejected.	
7	Pramod Dissanayake	pramoddissa@gmail.com	2020-10-23 15:00:00	Moragalla Estate	CO/GAL/0000107/2018	Galle	Galle Extension Centre	Head Galle	Land Verified. Membership Awarded.	
8	Kamila Wijepala96	kamila2009@gmail.com	2020-09-27 17:47:33	GalleTest Estate3	PR/GAL/0000384/2019	Galle	Galle Extension Centre	Head Galle	96 day(s) have elapsed responding	
9	Saman D Devapriya	sam.devpriya@yahoo.com	2020-09-22 18:52:04	Arya Estate	PR/GAL/0000111/2019	Galle	Galle Extension Centre	Head Galle	101 day(s) have elapsed responding	

Showing 1 to 9 of 9 entries

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Verify Tea Land for Member Registration

Home

Search

Tea Grower Support System

Logout

Head Galle

Member

Land Suitability

54

Dashboard > Member > New Registration

Member Profile

New Member

Name of the Tea Grower

Kapila Gamage

TGSS Membership No

TGSS000000180

Personal Details

Grower Category: Private

Gender: Male

NIC Number: 197012345671

Contact Details

Mobile No: 0771264567

Email: gamagekmc1@gmail.com

Postal Address: Test Address, Karapitiya.

Details of Tea Land

Land Name: GalleTest Estate2

Tea Board Registration No: PR/GAL/0000381/2019

Total Tea Extent (ha): 12.00

Total Extent (ha): 12.40

Geo Location of Tea Land

Administrative District: Galle

Divisional Secretariat Division: Karadeniya

Grama Niladhari Division: Walinguruketiya

Verify Tea Land - Link with Membership Account

* Required information

Inspection Details

Requested Date:

12/06/2020

Schedule Date:

12/22/2020

Inspection Date:

02/02/2021

Inspection Start Time: *

09:00 AM

Inspection End Time: *

12:51 AM

Verify Proof Documents: *

☐ National Identity Card
 ☐ Tea Board Registration Certificate

Details of Tea Cultivation

Please fill following details only, if you are Approving this request.

No of Division/Block(s): *

1

Record Keeping: *

☐ Systematic
 ☐ Adhoc
 ☐ None

Plant Type: *

☐ VP
 ☐ Seedlings
 ☐ Mix

Average Crop (kg/ha/yr): *

example: 2000

Remarks:

Enter your comments here. If any...

- I have met the Tea Grower and inspect the Tea Land.
- I have satisfied with the proof documents provided.

Approve

Reject Request

Please fill following details only, if you are Rejecting this request.

Reject Reason: *

Enter reject reason here.

Reject

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Version 1.0.0

Land Suitability Assessment

TGSS

Home

Search

Tea Grower Support System

Logout

Head Galle

Member

Land Suitability

Dashboard > Member > New Registration

19186

Land Suitability Assessment Request

Details of Request

Details of Land Block

Reference No: LS1220000086
Requested Date: 2020-12-27
Nature of Land Block: Bare Land Block
Extent (ha): 1.50
Reason for New Planting: New Investment

Photograph 1:

Photograph 2:

Tea Land Profile

Land Name: Talangaha Estate
TB Registration Number: CO/GAL/0000109/2018
Total Tea Extent (ha): 33.43
Total Extent (ha): 699.42
Administrative District: Galle
Divisional Secretariat Division: Yakkalamulla
Grama Niladhari Division: Nakiyadeniya North

Climate & Suitability for Tea

Sub Agro-Ecological Region: WL2a
Suitability Classification: Suitable to Cultivate Tea

Member Profile

Name: Hashan Dissanayake
TGSS Membership No: TGSS1020000191
Grower Category: Private

Contact Details

Mobile No: 0773334567
Email: hwdissa7611@yahoo.com
Postal Address: No 100 Kamburupitiya Road Akuressa

Assess Land Block to Cultivate Tea

Inspection Details

Scheduled Date:

01/14/2021

Inspection Date: *

02/02/2021

Inspection Start Time: *

02:07 PM

Inspection End Time: *

12:46 AM

Photographs of Land Block:

Photograph 1: *

Choose File

No file chosen

Photograph 2: *

Choose File

No file chosen

Upload photographs of land block depicting soil conditions. Only JPG, JPEG, PNG or GIF files are allowed with 800KB or lesser capacity.

Assessment of Limitations

Limiting Factor	Result	Degree of Limitation
Sub AER:	WL2a	None
Slope (%): *	exampl	Decision Pending
Soil Depth (cm): *	exampl	Decision Pending
Surface Rockiness (%): *	exampl	Decision Pending
Gravel Volume (%) in top 50 cm: *	exampl	Decision Pending

Check Limitations

Degree of Limitation:

Decision Pending

Based on TRI Advisory Circular No. 3/02 "Guidelines on Land Suitability Classification for Tea" issued in October 2002.

Suitability Class:

Decision Pending

Classification based on Degree of Limitation.

Approved Extent (ha): *

1.50

Recommendation: *

Enter your recommendation here.

Remarks:

Enter your comments here, if any...

I have visited the Tea Land.

I have assessed the Land Block according to suitability criteria.

Go Back

assess

Submit Report

Reset

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Section 2

Main Website

Home Page

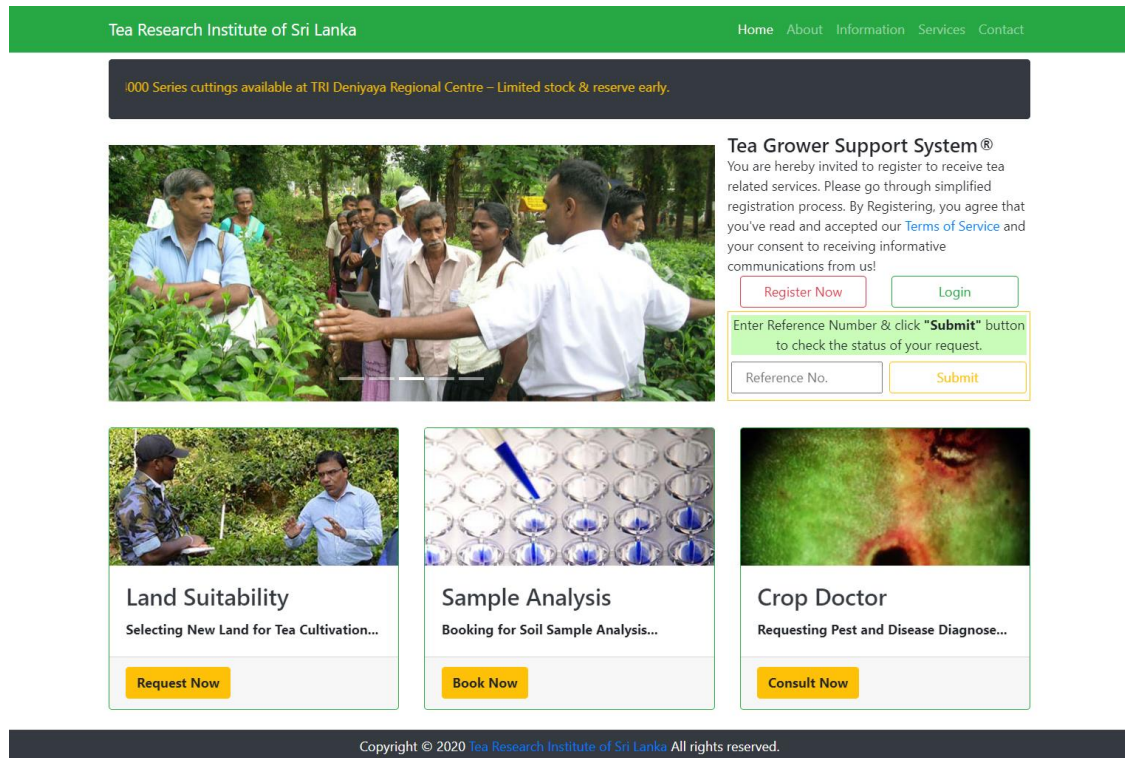


Figure C. 1 Home Page

The figure C.1 shows the home page of the main site that will be enabled for all user categories. It provides the menu bar and links to navigate easily to user through the website. In addition, member registration and login links are also included. It contains welcome message on the right with slideshow. Finally, footer of the page consists of contact details and logo of the client and copyright statement.

Home Page menu

The menu items helps user to scroll through whole web site.

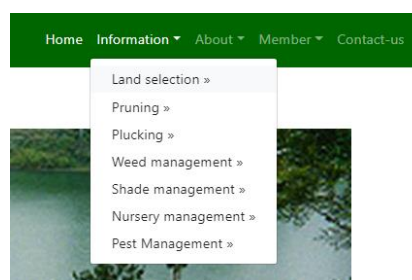


Figure C. 2 Home Page Menu

Tea Information

View Tea Information

In menu user selects Information > Plucking

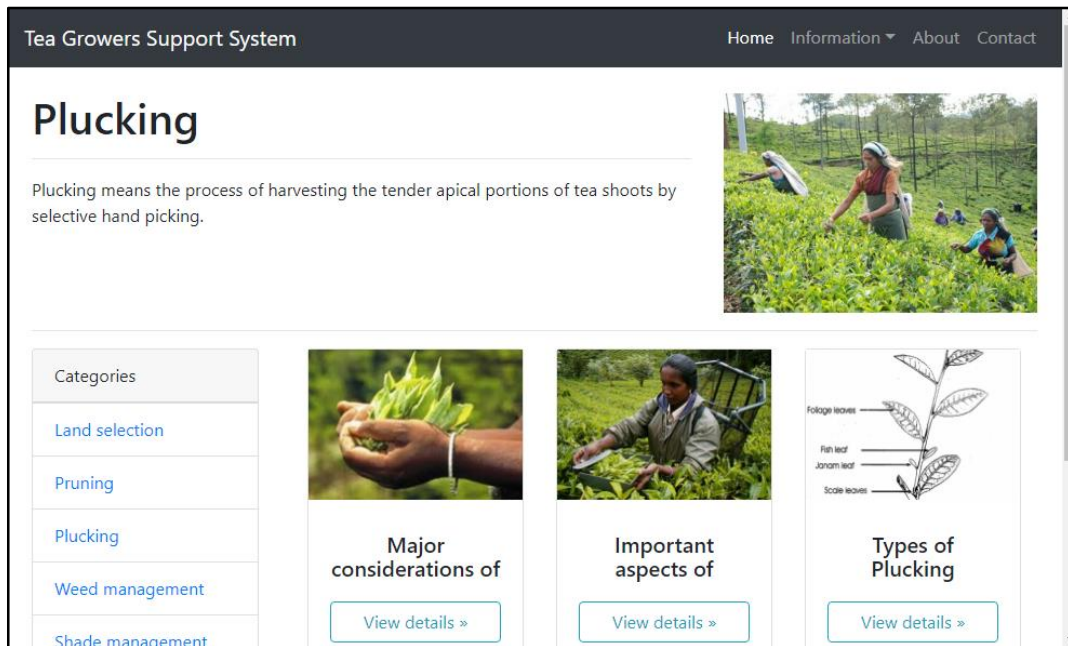


Figure C. 3 Tea Information Category Page

Appendix D - Management Reports

The system allows the administrators and Module Administrators to generate reports. Administrator can obtain list of members who registered to the system with their status.

Transaction Reports

Sample Analysis Report

	Tea Grower Support System Tea Research Institute of Sri Lanka <u>Talawakelle</u> Telephone: 052 2258201 Galle: Laboratory: 09122222564					
Report Name: Soil sample Report		Date: 30.12.2020				
Member Name: <u>Hashan Dissanayake</u> TGSS Registration No. TGSS1020000191						
Tea Land Name: <u>Talangaha Estate</u> Tea Land Registration No. CO/GAL/0000109/2018						
Date of Sample: 12.12.2020						
Sample Ref No.	Sample	P (ppm)	K (ppm)	Mg (ppm)	pH	C %
LS1220000086	Soil	171A	256A	115A	4.33A	4.46L
H - High A - within acceptable level M - Marginally low L = Low						
Laboratory ID: GL						
This is a computer generated report. No signature required,						
Created By: analyst.galle.lk		Printed on: 30.01.2021 13:50			Page 1 of 1	

Management Reports

Monthly Member Registration Report

This report representing daily count of successful member registration for given month.

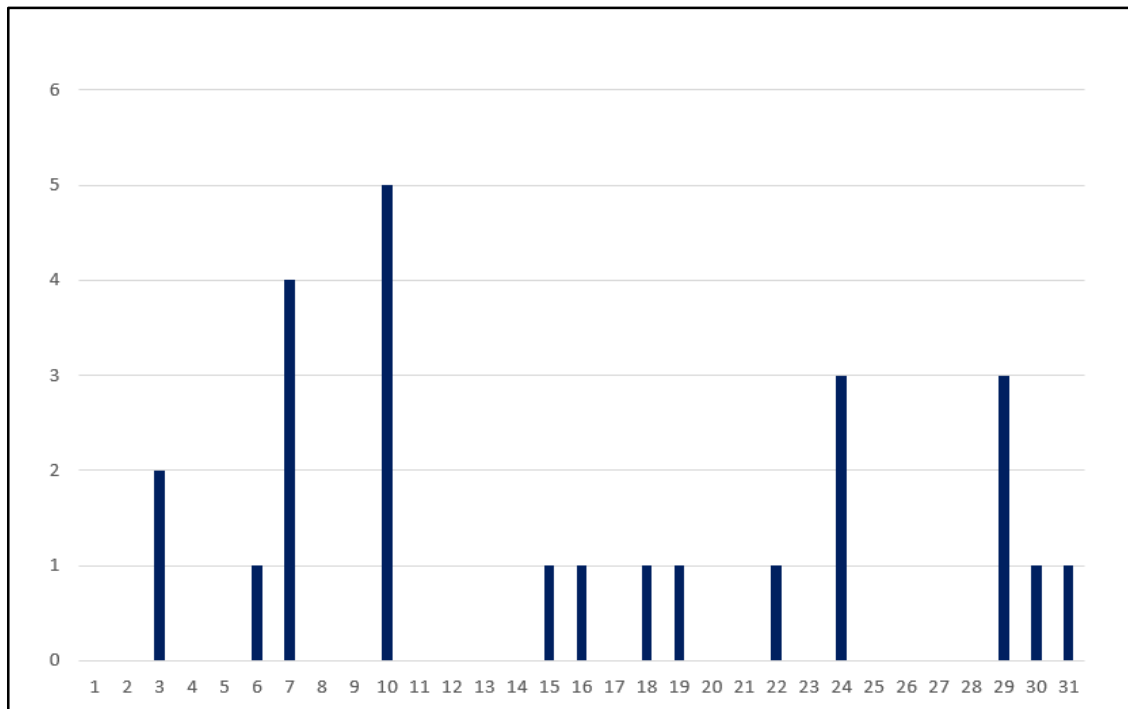


Figure D. 1 Monthly Member Registration Report

Report of Service Requests - This Month

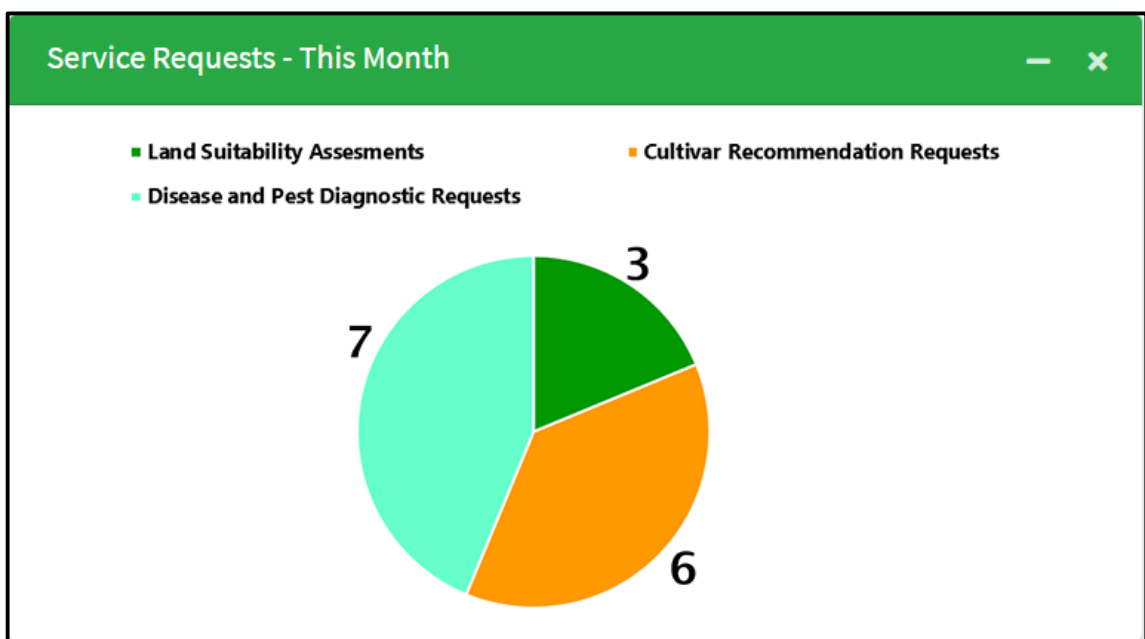
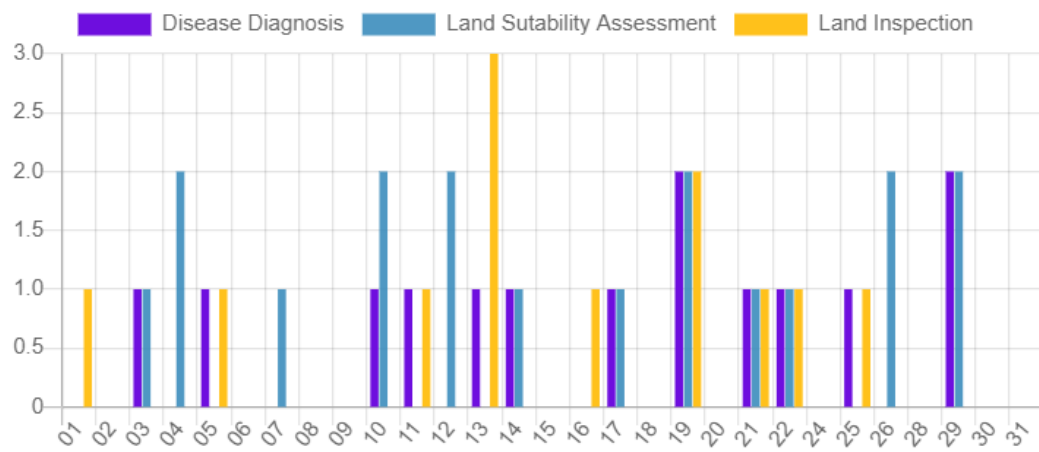


Figure D. 2 Service Requests – This month

Report of Daily Attended Request Summary

Daily Attended Request Summary - December 2021



Appendix E – Test Results

General Evaluation

ID	Test Description	Testing Procedure	Actual Output	Status
General Evaluation – Test Plan				
1	Page Title	Appropriateness – Main site	Appropriate	Pass
2	Font size	Appropriateness – Main site	Appropriate	Pass
3	Font type	Appropriateness – Main site	Appropriate	Pass
4	Spelling of page content	Check for spelling mistakes - Main site	No spelling errors	Pass
5	Grammar of page content	Check for grammar mistakes - Main site	No grammar errors	Pass
6	Page navigation	Click on links/ buttons - Main site	Redirects to relevant page	Pass
7	Relevancy of images	Appropriateness – Main site	Appropriate	Pass
9	Responsiveness of the interfaces	Resizing interface - 19-inch wide screen monitor	Appropriate	Pass
10		Resizing interface - 17-inch standard square shape monitor	Appropriate	Pass
11		Resizing interface - 15.6-inch wide laptop screen	Appropriate	Pass
12		Resizing interface - 8-inch tablet screen	Appropriate	Pass
13		Resizing interface - 5.5-inch smartphone screen	Appropriate	Fail
Common Functions – Test Plan				
14	Login in to the system	Submit login with empty field	An error message will appear	Pass
15	Login in to the system (Negative)	Enter an invalid user name or password	An error message/s will appear	Pass
16	Login in to the system (Positive)	Enter valid user name and password	Successfully logged in to the	Pass

			system and redirects to dashboard	
17	Display user name after login	Logged in to the system with proper authentication	Display first name and last name on the top status bar	Pass
18	Display profile picture after login	Logged in to the system with proper authentication	Display profile picture on the top status bar	Pass
19	Privileged module access according to user type	Logged in to the system with proper authentication	Display assigned modules on the left navigation pane of the dashboard	Pass
20	Logout form the system	Click on the logout button	The user logged out from the system and redirect to the login page	Pass
User and Module Management – Test Plan				
21	Register an officer	Fill all the * marked inputs parameters of the officer who is not in the system	Register the officer and display success message	Pass
22	Validate empty filed of register officer form	User hits submits form with all mandatory fields empty	Display error messages	Pass
23	Check the given phone no has valid length	Enter phone with invalid length	Display error message	Pass
24	Invalid first name given	Enter invalid characters	Display error message	Pass
25	Invalid email given as user name	Validate email address is in correct format	Display error message	Pass
26	Insert too large capacity file as profile image	Validate file size of the selected profile image in allowed capacity	Display error message	Pass

27	Upload different type of file as profile image	Validate if the image file is an actual image or fake	Display error message	Pass
28	Enter username	Check whether username already taken or not in the database	Display error message	Pass
29	View profile of officer already in the system	Open View user menu and click on View icon of the selected officer	Display profile of the officer	Pass
30	Modify details of registered officer	Edit relevant field(s) and click Update button	Update the database with new details	Pass
31	Delete registered officer profile	Select relevant officer record and click 'Delete' icon	Prompt confirmation window, if user selects 'Yes' delete the officer	Pass
32	Assign new role to user	Select user name, module name then click submit button	Role assigned to user account and display success message	Pass
33	Assign already assigned role to user	Select user name, module name then click submit button	Display error message	Pass

Table E. 1 All Test case of the system

Glossary

AJAX	Stands for Asynchronous JavaScript and XML. This technique is used to make the interface more interactive by checking any data input field without reloading the page with server connectivity
Jquery	Scripting technology which uses for client side validation

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yield data iii